

Explainable Dual-Head CNN–BiLSTM–Attention Prognostics for Industrial Bearing Health Monitoring

A beginner-friendly learning guide, viva question bank and speaking script

Primary notebook	final_project/final_bearing_prognostics_review.ipynb (93 cells: 44 Markdown + 49 code; all code cells carry saved outputs, 0 error outputs)
Dataset	PRONOSTIA / IEEE PHM 2012 Prognostic Challenge bearings (FEMTO-ST, Besançon)
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Rule used to write this guide	Every number is taken from saved notebook outputs or saved artifact files (CSV/JSON). Nothing was re-run, re-trained or modified. Anything not found is labelled unavailable .
Cell-index convention	Zero-based , exactly as stored in the .ipynb file: the first cell (title) is [0] and the last (references) is [92]. Notebook “[n]” execution counters (1–49) are a different thing and are written as “exec n”.

Read this first (three honest headlines you must be able to say aloud).

1. On 11 unseen test bearings the proposed model recognises an *imminent* failure better than chance (mean detection AUC 0.749 vs 0.50 for a constant predictor), but it is **statistically tied** with a plain CNN and a dual-head model without attention, and it is **not** the most accurate model for minute-level RUL (Gradient Boosting is lower).
2. The hypothesis “a useful warning for every bearing” is **NOT supported**: 3 of 11 test bearings get a useful warning (alarm active at the end and sustained lead ≥ 10 min).
3. Attention is **functionally weak** (replacing it by uniform weights changes the RUL prediction by only 1.4 min on average) and is not a causal explanation. We claim *inspectability*, not *causal explanation*.

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1. Project at a glance

1.1 Title, problem and motivation

Title. Explainable Dual-Head CNN–BiLSTM–Attention Prognostics for Industrial Bearing Health Monitoring (notebook cell [0]).

Problem statement (cell [91], item 5). Given the recent vibration history of a bearing, estimate its **remaining useful life (RUL)** and its **health stage** (Normal / Warning / Critical), for bearings that were *never seen* during training.

Motivation (cell [91], item 1). Bearings fail suddenly and unplanned stops are expensive. If the vibration data can tell a maintenance engineer that a bearing is close to failure, maintenance can be planned instead of reacting to a breakdown.

1.2 Objectives (the six research questions, cell [1])

RQ	Question (short)
RQ1 Data	What does the dataset really contain, and are the facts we rely on (lifetimes, end of life, file formats) consistent?
RQ2 Formulation	Which prognostic target (raw/capped/normalised RUL, stage classes, degradation detection, multi-task) is scientifically defensible?
RQ3 Generalisation	How well does vibration predict remaining life and health stage of a bearing never seen (bearing-level validation, no leakage)?
RQ4 Architecture	Does each component (CNN, BiLSTM, attention, second head) measurably contribute versus simple baselines?
RQ5 Explainability	What can attention weights and feature importance truthfully tell us?
RQ6 Limits	For which bearings is early warning physically possible, and for which is failure too abrupt?

1.3 Proposed solution in one paragraph

Every 10 seconds the rig records a 0.1-second vibration snapshot (2560 samples, two accelerometers). Each snapshot is converted into **34 numbers** (17 features per channel: time-domain and spectral). The model looks at the **last 16 snapshots** (a 16×34 table). A **CNN** finds short local patterns, a **BiLSTM** follows the trend inside the window, **attention** weights the 16 time steps, and **two heads** output (i) the remaining life in minutes (capped at 120) and (ii) a three-class stage. Evaluation is by **whole bearings** only (cells [34]–[35]).

1.4 Contributions that ARE supported by the saved evidence

#	Contribution	Evidence (cell / artifact)
1	A leakage-controlled, bearing-level protocol with 20 automated checks, all passed (disjoint splits, scaler on training bearings only, poison test, truncation/causality test, target-shuffle test, frozen configuration).	[32]; audits/FINAL_AUDIT_automated.md
2	Quantified proof that a random window split inflates results: MAE 1.5 min / AUC 1.00 (random) versus MAE 42.2 min / AUC 0.70 (bearing-level), same model, same bearings.	[35]
3	A programmatic dataset audit that found a real anomaly: Bearing1_4 in Full_Test_Set runs 2551 s past its official end of life (handled by truncation); the 20 g criterion is not visible in every snapshot.	[7], [9]
4	An honest comparison of 11 models (constant, Ridge, Random Forest, Gradient Boosting, CNN, LSTM, CNN+LSTM, CNN+BiLSTM, +attention, dual-head variants, proposed) under identical protocol, with paired-bootstrap ablations and the official PHM-2012 score.	[45], [51], [69], [73], [83]
5	On the unseen test bearings the proposed model detects imminent failure better than chance (AUC mean over horizons 0.749; 0.759 at 10 min) and has a lower actionable-region MAE than a constant predictor (38.1 vs 56.4 min).	[69], [76], [91]
6	A saved, deployable pipeline (scaler + Keras model + feature config) and a causal one-window demo.	[68], [79], [80]
7	Explainability tools that are <i>measured</i> rather than asserted (uniform-attention functional test, permutation importance).	[59]–[62], [74]

Not supported (do not claim): accurate minute-level RUL prediction; a long warning for every bearing; that the proposed architecture is more accurate than simple baselines; causal explanation by attention; comparability with PHM-2012 challenge leaderboards (the predictor is capped at 120 min and the official score uses only 11 predictions). Source: cells [84], [86], [87].

2. Fundamentals explained from zero

Each topic has a plain-English explanation first and a short technical note after it.

2.1 What an industrial bearing does

A bearing lets a shaft rotate with very little friction while carrying a load. It is made of an inner ring, an outer ring and rolling elements (balls or rollers) held by a cage. Motors, pumps, fans and gearboxes all depend on bearings.

Technical / project note: In the dataset the rig runs bearings at 1800 rpm / 4000 N, 1650 rpm / 4200 N or 1500 rpm / 5000 N (cell [2], fact 6).

2.2 Bearing degradation and failure

Over time, the surfaces wear and crack. Small defects make small impacts every time a ball rolls over them; the impacts grow, vibration rises and eventually the bearing fails. Real bearings do not follow one pattern: some deteriorate slowly for hours, others look healthy until minutes before failure.

Technical / project note: Cell [15] states this directly for the six learning bearings: Bearing1_1 rises slowly, Bearing3_1, 3_2 and 1_2 stay flat and rise only in the last few minutes (final phase 3.8–9 min, table in cell [14]).

2.3 Vibration monitoring

An accelerometer glued to the bearing housing measures how strongly it shakes, in units of g (gravitational acceleration). Healthy bearings give a low, noise-like signal; damaged ones give sharp bursts.

Technical / project note: Here: two accelerometers (horizontal and vertical, 90° apart), 25.6 kHz sampling, a 0.1 s snapshot (2560 samples) every 10 s (cell [2], facts 7–10). Healthy snapshot peaks 1.7 g; the end-of-life snapshot exceeds 20 g (cells [11], [12]).

2.4 Predictive maintenance

Instead of fixing a machine on a fixed schedule or after it breaks, we use data to estimate its condition and repair it when needed. This saves cost and avoids surprise breakdowns.

Technical / project note: This project is a data-driven prognostics study; it does not control a real plant.

2.5 Remaining useful life (RUL)

RUL is the time left before the bearing fails: $RUL(t) = \text{failure time} - t$. At the start of a 400-minute test the RUL is 400 minutes; at the end it is 0.

Technical / project note: Failure here is not defined by thresholding the snapshots (the 20 g rule is not visible in every 0.1-s snapshot) but by the **official recording duration** (learning bearings) or **truncation point + official RUL** (test bearings); cells [9], [14].

2.6 Feature engineering

Raw 2560-number waveforms are hard for a small dataset to learn from. We compress each waveform into a few meaningful numbers: overall energy (RMS), peak, “spikiness” (kurtosis, crest factor), and how the energy is spread over frequencies (spectral centroid, band energies).

Technical / project note: 17 features per channel × 2 channels = 34 (cell [21]). Heavy-tailed ones are log-compressed with *fixed* (non-fitted) transforms, so nothing is learned from test data.

2.7 Deep learning

A neural network is a function with many adjustable numbers (weights). Training adjusts the weights to reduce an error (loss) on example data. With very few examples (here 6 learning bearings), a big network can simply memorise them.

Technical / project note: Cell [42] shows exactly this: training loss falls to nearly zero while validation loss on an unseen bearing rises.

2.8 CNN, BiLSTM, attention, dual head

CNN (1-D convolution): a small sliding filter that detects local patterns across neighbouring time steps. **BiLSTM**: a memory-based layer (LSTM) that reads the sequence in both directions to capture trends; “bidirectional” is legitimate here because the whole window is in the past. **Attention**: learns weights α_t that say how much each of the 16 time steps matters for the answer. **Dual head**: one shared representation feeds two outputs: a number (RUL) and a class (stage).

Technical / project note: Details and the verified layer table are in Section 5. The measured contribution of each component is in Sections 6 and 11; in short: not reliably shown to help.

2.9 Training, validation and testing

Training data is what the model learns from. **Validation** data is held back during training to decide when to stop (early stopping) and to choose between designs. **Test** data is touched once at the very end to estimate how it will behave on a new machine.

Technical / project note: Here the unit is the **bearing**, not a window: 6 learning bearings are rotated through leave-one-bearing-out folds (4 train, 1 validate, 1 held out); the 11 test bearings are scored once after the design is frozen (cells [34], [67]).

2.10 Data leakage and generalisation

Leakage means information from the data you evaluate on has secretly reached training. Generalisation means doing well on genuinely new data. Neighbouring windows of one bearing are nearly identical copies of each other, so a random window split lets the model “see” its test windows during training and gives fake-good scores.

Technical / project note: Cell [35] measures it: same model, random split MAE 1.5 min / AUC 1.00; bearing split MAE 42.2 min / AUC 0.70.

2.11 Explainability

Explainable AI tries to show *why* a model gave its answer. Two common tools: attention weights (where in the input the model looked) and permutation importance (how much worse predictions become if one feature is scrambled).

Technical / project note: Both are **descriptive**: they describe the model, not the physical cause of the failure (cell [58], [62]).

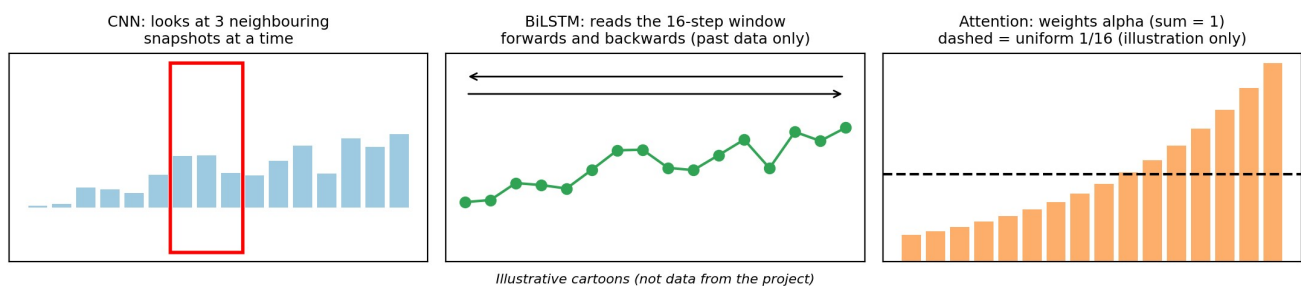
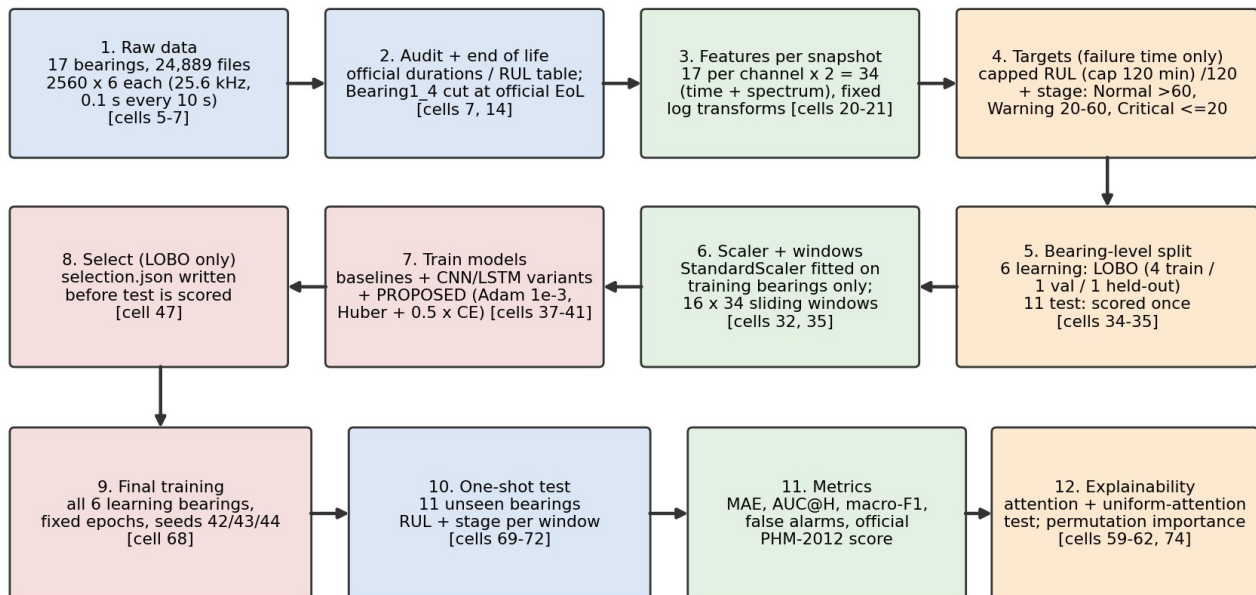


Figure A. Illustrative cartoons of the three building blocks (drawn for this guide; not project data).

3. Complete end-to-end workflow

The diagram below was drawn for this guide from the verified implementation (`src/config.py`, `data.py`, `features.py`, `models.py`, `metrics.py`, `experiment.py` and the notebook code cells). Box labels give the notebook cell where each step is executed.



Schematic drawn for this guide from the verified implementation (`src/*.py` and notebook code cells); it is not an image from the notebook. Colours only group the boxes visually (data / features / targets and splits / training / evaluation).

Figure B. Implemented pipeline (schematic created for this guide).

3.1 Step by step in plain English

Step	What happens	Where
1 Load	Every raw file (24,889 snapshots × 2560 samples × 2 channels = 127,431,680 samples) is read once; features are computed from it.	[5], [7]
2 Audit	Check shapes, NaN/Inf (none), time stamps, where 20 g is crossed, and whether Test_set matches Full_Test_Set (33/33 numerically identical).	[7]
3 End of life	Learning bearings: last recorded snapshot. Test bearings: truncation point + official RUL. Snapshots after that are discarded (24,633 of 24,889 kept).	[14]; <code>src/data.py</code>
4 Features	34 features per snapshot; fixed log / signed-log transforms.	[20], [21]
5 Targets	Capped RUL (120 min cap, divided by 120 → 0...1) and a 3-class stage, both from the failure time only.	[25], [29]
6 Split	6 learning bearings in 6 LOBO folds; 11 test bearings held back for one final scoring.	[34], [35]
7 Scale + window	StandardScaler fitted on the training bearings of each fold; sliding windows of 16 snapshots built inside each bearing, labelled with the last snapshot's targets.	[32], <code>make_windows</code>
8 Train + compare	11 model types trained in the same folds with 2 seeds (LOBO) and 3 seeds (final).	[37], [41]
9 Select	Selection rules applied to LOBO results only; <code>selection.json</code> written before the test is scored.	[47]
10 Final test	Retrain on all 6 learning bearings with fixed epochs, score 16,934 windows of the 11 test bearings once.	[68]–[72]
11 Explain	Attention profile, uniform-attention test, permutation importance.	[59]–[62], [74]

4. Dataset and preprocessing

4.1 The dataset (verified from cells [2] and [7])

Item	Value	Verified how
Source	PRONOSTIA platform, FEMTO-ST Institute, Besançon, France; IEEE PHM 2012 Prognostic Challenge	Literature facts, cell [2] (from the challenge document and platform paper)
Bearings	17 run-to-failure experiments: 6 learning + 11 test	Assertion in cell [7]
Operating conditions	C1 1800 rpm/4000 N; C2 1650 rpm/4200 N; C3 1500 rpm/5000 N	Cell [7] table columns rpm, load_N
Snapshots (acc files)	24,889 total; each 2560 rows × 6 columns; no NaN/Inf	Cell [7] output
Sampling	25.6 kHz; 2560 samples = 0.1 s; one snapshot every 10 s	Cell [11]: 100 ms, 39 μs spacing check passed
Channels	Horizontal and vertical acceleration (g)	Cell [11] column table
Lifetimes	38 to 467 min (factor ≈ 12). Min/median/max: C1 145.0/395.7/467.0; C2 38.2/132.7/385.0; C3 72.2/85.7/272.7 min	Cell [8] markdown output (computed over all recorded files, so Bearing1_4 shows 237.8 min; after truncation at its official end of life it is 195.2 min, cell [74])
Delimiter anomaly	Only Bearing1_4 (Full_Test_Set) uses ';' instead of ',' (1,428 files)	Cell [7]

4.2 The 34 features (cell [21]; artifact feature_documentation.csv)

Feature (per channel)	Formula	Transform	Why it may indicate degradation (notebook text)
rms	$\sqrt{\text{mean}(x^2)}$	log10	Overall vibration energy; grows as defects enlarge and friction/impacts increase.
peak	$\max x $	log10	Largest instantaneous shock; sensitive to single strong impacts (also to noise spikes).
p2p	$\max(x) - \min(x)$	log10	Total excursion; similar to peak but based on the signed range.
kurtosis	$\text{mean}((x-\mu)^4)/\sigma^4 - 3$	signed log1p	Impulsiveness. About 0 for smooth Gaussian vibration, rises when sharp impacts appear (spalls, cracks).
skewness	$\text{mean}((x-\mu)^3)/\sigma^3$	signed log1p	Asymmetry of the amplitude distribution; one-sided impacts change it.
crest_factor	peak/rms	log10	Peak-to-average ratio; rises early with isolated impacts, falls when damage becomes continuous.
shape_factor	$\text{rms}/\text{mean} x $	log10	Waveform shape; changes when the amplitude distribution departs from Gaussian.
impulse_factor	$\text{peak}/\text{mean} x $	log10	Sharpness of impulses relative to the mean level.
margin_factor	$\text{peak}/(\text{mean}(\sqrt{ x }))^2$	log10	Peak relative to a low-order mean; emphasises rare large peaks (classical incipient-fault indicator).
spec_energy	$\sum(P_k), P = \text{FFT}(x-\mu) ^2/N^2$	log10	Total spectral power (Parseval: equals variance); tracks overall energy growth.
spec_centroid	$\sum(f_k P_k)/\sum(P_k)$	none	Centre of gravity of the power spectrum; moves when high-frequency resonances are excited.
spec_spread	$\sqrt{\sum((f_k - f_c)^2 P_k)/\sum(P_k)}$	none	Bandwidth of the spectrum; broadens with damage.
spec_peak_freq	$\text{argmax}_f P(f)$	none	Dominant frequency (shaft harmonics vs structural resonance).
band_0_1k	$\sum(P_k, 0-1 \text{ kHz})/\sum(P_k)$	$\log_{10}(x+1e-6)$	Fraction of power at low frequency (shaft rotation harmonics; shaft speed is 25-30 Hz).
band_1_3k	$\sum(P_k, 1-3 \text{ kHz})/\sum(P_k)$	$\log_{10}(x+1e-6)$	Mid band.
band_3_6k	$\sum(P_k, 3-6 \text{ kHz})/\sum(P_k)$	$\log_{10}(x+1e-6)$	Mid-high band.
band_6_12k	$\sum(P_k, 6-12.8 \text{ kHz})/\sum(P_k)$	$\log_{10}(x+1e-6)$	High band, where impact-excited resonances appear.

Counts: 9 time-domain + 4 spectral-shape + 4 band-energy features = 17 per channel; $\times 2$ channels (horiz_, vert_) = **34**. Transforms: log10 on 16, $\log_{10}(x+1e-6)$ on 8, signed log1p on 4 (output of cell [21]). The standard deviation was intentionally excluded (it duplicates RMS). Every feature uses **one snapshot only**, so it cannot depend on the future (verified by check 17 in cell [32]).

Two equations used by the features (rendered from the formulas in cell [21]):

$$x_{\text{rms}} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} \quad \text{kurtosis} = \frac{\frac{1}{n} \sum (x_i - \mu)^4}{\sigma^4} - 3$$

RMS measures overall vibration energy; kurtosis measures “spikiness” (≈ 0 for smooth Gaussian vibration, larger when sharp impacts appear). They are the best-known bearing-condition features, but cell [19] shows that **no single feature is a consistent health indicator across bearings**: only 2 of 34 have a consistent whole-life trend (and the most consistent, the 3–6 kHz band fractions, *decrease* with damage). 18 of 34 features shift by more than one robust standard deviation in the last 60 minutes.

4.3 Labels: how RUL, capped RUL and stage are defined (cells [25], [29]; src/data.py)

End of life (EoL). Learning bearings: time of the last recorded snapshot (official recording duration). Test bearings: (truncation snapshot – 1) × 10 s + **official actual RUL** from the challenge table. Reason (cells [7], [9]): a snapshot covers only 0.1 s of every 10 s, so the 20 g criterion is not observable in every snapshot (e.g. Bearing1_5 peaks at 14.1 g, 2_4 at 8.8 g), while others exceed 20 g long before the official EoL (e.g. Bearing1_2 at 14.6 % of life).

RUL in minutes = (EoL – snapshot time) / 60. **Capped RUL** = min(RUL, 120 min); the network target is capped RUL / 120 between 0 and 1 (sigmoid output).

$$y_t = \frac{\min(\text{RUL}_t, 120 \text{ min})}{120 \text{ min}}$$

Stage (3 classes) from the true RUL only: Normal if RUL > 60 min; Warning if 20 < RUL ≤ 60; Critical if RUL ≤ 20 (stage_from_rul_min, config STAGE_WARNING_MIN=60, STAGE_CRITICAL_MIN=20). **The cap (120) and the boundaries (20/60) are design choices fixed a priori**, not tuned; cell [30] shows that the conclusion does not depend on the cap (60/120/240).

Why not the other targets? (cell [25]–[29]) The Random-Forest probe on the six learning bearings did **not** clearly beat a constant for minute-level RUL (uncapped: 103.3 vs 97.2 min, better in 3/6 folds; capped: 42.2 vs 38.8 min, 2/6 folds). Life-fraction (B) looked better (0.20 vs 0.25) only because it needs the bearing's total lifetime = future information, so it is not deployable and was rejected. Stage classification beat the majority class in 5/6 folds (macro-F1 0.43 vs 0.25).

4.4 Windows, scaling and the bearing-level split

Sequence construction (make_windows): a window is the 16 most recent snapshots (k ... k+15) *within one bearing*; it is labelled with the targets of its **last** snapshot. The first window of every bearing ends at snapshot 15, so there is no padding across bearings (check in cell [32]). 16 snapshots span 150 s between first and last (the notebook calls it 160 s of history). Windows overlap by 15 of 16 snapshots.

Scaling. A StandardScaler is fitted on the training bearings of each fold only (and on all 6 learning bearings for the final model); it is applied to validation, held-out and test windows. Check 1 in cell [32] asserts that the scaler means equal the training-bearing statistics and differ from all-data statistics.

LOBO folds (output of cell [35]):

Fold	Held-out (scored)	Validation (early stopping only)	Training
0	Bearing1_1	Bearing1_2	2_1, 2_2, 3_1, 3_2
1	Bearing1_2	Bearing2_1	1_1, 2_2, 3_1, 3_2
2	Bearing2_1	Bearing2_2	1_1, 1_2, 3_1, 3_2
3	Bearing2_2	Bearing3_1	1_1, 1_2, 2_1, 3_2
4	Bearing3_1	Bearing3_2	1_1, 1_2, 2_1, 2_2
5	Bearing3_2	Bearing1_1	1_2, 2_1, 2_2, 3_1

Final test bearings (11, never in any training or selection step): Bearing1_3, 1_4, 1_5, 1_6, 1_7, 2_3, 2_4, 2_5, 2_6, 2_7, 3_3 (config. TEST). **Final training set:** all 6 learning bearings (7,444 windows); scored: 16,934 test windows (cell [68]).

Why split by bearing (explain this aloud)

- Consecutive windows of one bearing share 15 of 16 snapshots, so they are almost duplicates. A random split puts near-copies of test windows into training; the model then only has to “recognise” a bearing it has already seen.
- In reality the model must handle a **new** machine. The bearing is therefore the independent unit (the challenge itself separates whole bearings).
- Measured in cell [35] with the same Random Forest and bearings: random window split → MAE **1.5 min**, AUC **1.00**; bearing-level LOBO → MAE **42.2 min**, AUC **0.70**.

4.5 Leakage: what was verified and the caveats

I did not accept the Markdown claim; I read the implementation (`src/data.py`, `leakage.py`, `experiment.py`) and the saved output of cell [32]:

Leakage route	Evidence in code / output	Verdict
Train/test bearings overlap	<code>assert_disjoint</code> and <code>check_splits</code> : folds disjoint, learning-only, each learning bearing held out once; no TEST bearing in the fold-0 training set is asserted. Output [32] row 0 True.	Verified
Scaler fitted on more than training data	<code>fit_scaler(feats, table, f['train'])</code> in <code>run_lobo</code> (validation bearing excluded); <code>check_scaler_train_only</code> passes for all 6 folds.	Verified
Held-out data influencing training	Poison test: held-out features replaced by random numbers $\times 10^3$ and targets zeroed $\rightarrow$ training windows, scaler and metadata bit-identical (6/6 folds True).	Verified
Future data in an input window	Truncation test on Bearing1_1, 2_4, 1_3 at 3 cut points: past windows unchanged. Window label = last snapshot. Features from a single snapshot (check 17).	Verified
Targets using vibration hindsight	Feature-shuffle test: targets identical after shuffling all features (check 18). No onset or life-fraction label used for the final models.	Verified
Test used for decisions	<code>selection.json</code> written before test predictions (timestamps in FINAL_AUDIT_automated.md check 6); config snapshot re-asserted; cell [68] refuses to run twice per session.	Verified (by code + audit file)

Caveats you should state honestly (they are in the notebook or visible in the source):

- **Prior exposure.** Earlier experiments (v1/v2) looked at test-bearing curves; the *choice* of the capped-RUL + stage formulation was motivated by those results (cells [3], [84]; FINAL_AUDIT.md). The test set is therefore not perfectly virgin.
- **Test labels need the official RUL table** for end of life (evaluation only; they never enter training).
- **Code detail:** in `src/data.py` the line inside `fit_scaler` `assert set(train_bearings) <= set(C.LEARNING)` or `True` can never fail. The learning-only property is instead enforced by `check_splits` and `check_scaler_train_only` in cell [32], so it does not change the conclusion, but you should not point to that assert as evidence.
- **Few bearings:** condition 3 has only 2 learning bearings. From the fold table: in fold 4 (held-out Bearing3_1, validation 3_2) **no condition-3 bearing is in the training set**; folds 3 and 5 have only one.
- Passing the checks shows the *pipeline* does not leak; it cannot remove selection effects (choosing models by LOBO results, tuning nothing on test is a procedure, not a guarantee).

5. Model architecture

Everything below is taken from `src/models.py` (`build_nn("proposed")`) and the layer table printed by cell [39].

5.1 Layer-by-layer table (output of cell [39])

Layer	Type	Output shape	Params	Purpose
sequence_input	Input	(None, 16, 34)	0	16 snapshots × 34 features
conv1	Conv1D(64, k=3, ReLU, same)	(None, 16, 64)	6,592	local feature combinations over 3 neighbouring snapshots
bn1 / drop1	BatchNorm / Dropout 0.2	(None, 16, 64)	256 / 0	stabilise training / regularise
conv2	Conv1D(64, k=3, ReLU, same)	(None, 16, 64)	12,352	second local layer
bn2 / drop2	BatchNorm / Dropout 0.2	(None, 16, 64)	256 / 0	as above
bilstm	Bidirectional LSTM(64) + return_sequences	(None, 16, 128)	66,048	trend/order in both directions of the past window
drop_bilstm	Dropout 0.2	(None, 16, 128)	0	regularise
temporal_attention	custom TemporalAttention	[(None,128), (None,16,1)]	16,640	weights α_t of the 16 steps; returns context vector c and α
shared_dense / drop_shared	Dense 64 ReLU / Dropout 0.2	(None, 64)	8,256 / 0	shared representation for both heads
rul_dense → rul_output	Dense 32 ReLU → Dense 1 sigmoid	(None, 1)	2,080 + 33	capped RUL / 120 min in [0,1]
risk_dense → risk_output	Dense 32 ReLU → Dense 3 softmax	(None, 3)	2,080 + 99	probabilities of Normal / Warning / Critical

Total parameters: 114,692 (cell [39]; Keras `count_params`, which includes batch-norm moving statistics).

Derived by me, not printed: the 4 batch-norm moving-mean/variance vectors (2 layers × 128) are non-trainable, so trainable $\approx 114,692 - 256 = 114,436$ (this differs from the older “113,924” figure that belongs to the earlier v1 model, see Appendix D).

Parameter counts of all neural variants (cell [39]): CNN 25,729; LSTM 31,617; CNN+LSTM 58,753; CNN+BiLSTM 95,873; CNN+BiLSTM+Attention (single head) 112,513; CNN+BiLSTM dual head (no attention) 98,052; **proposed 114,692**. Cell [39] itself warns that this is a lot of capacity for only 6 learning bearings.

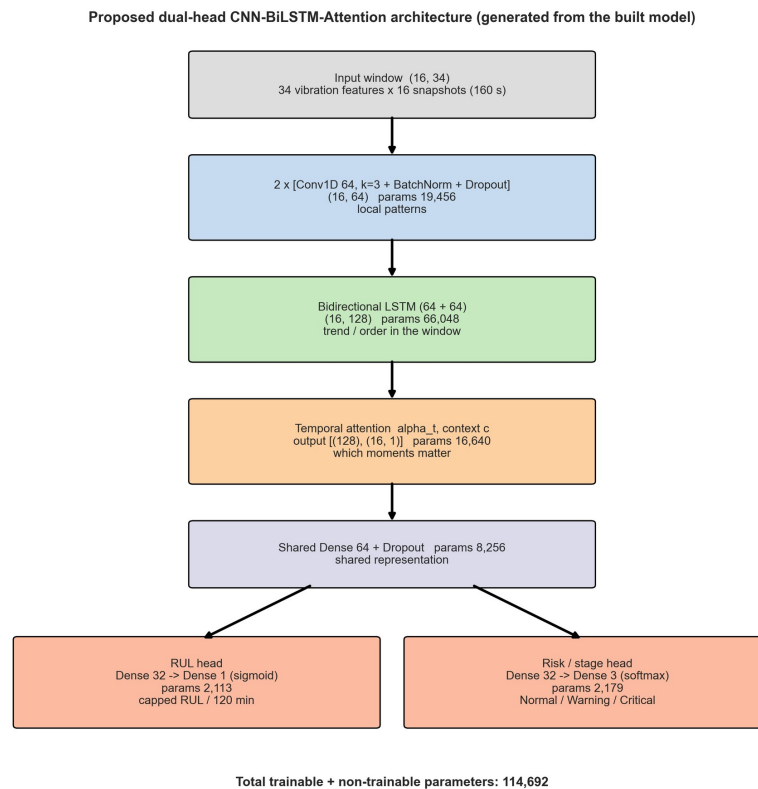


Figure G1. Architecture diagram generated by the notebook from the built Keras model. (Source: notebook cell [39]; saved file final_project/figures/fig_architecture.png)

What is shown	The data flow from the 16×34 input through two Conv1D blocks, the BiLSTM, temporal attention, a shared dense layer and two output heads, with output shapes and parameter counts taken from the model.
How to read it	Read top to bottom. Each box lists the layer, its output shape, its parameter count and its role. The final box gives the total parameter count.
What the result shows	The model is small by deep-learning standards (~115 k parameters) but large relative to 6 independent training histories; the BiLSTM (66 k) holds the most parameters.
Reasonable conclusion	This is the actual implemented architecture (generated from the model, not hand-drawn), so you can describe it confidently.
Do NOT conclude	That every block is necessary: ablations (Section 6.6) did not show a robust benefit for attention or the second head.

5.2 Tensor flow in words

- Input **(16, 34)**: 16 snapshots, 34 scaled features.
- Two Conv1D (64 filters, kernel 3, padding 'same') keep the 16 time steps and produce **(16, 64)**: each step now describes a mix of its 3 neighbours.
- BiLSTM(64) in two directions gives **(16, 128)**: one 128-number description per time step.
- Attention scores each of the 16 steps, softmax turns the scores into weights α (sum = 1), and the weighted sum gives one **128-number context vector**.
- Dense(64) $\rightarrow$ two heads. **RUL head**: Dense(32) $\rightarrow$ Dense(1, sigmoid) $\rightarrow$ multiply by 120 min. **Stage head**: Dense(32) $\rightarrow$ Dense(3, softmax).

$$\alpha_t = \text{softmax}_t(u^\top \tanh(Wh_t + b)), \quad c = \sum_{t=1}^{16} \alpha_t h_t$$

This is additive (Bahdanau-style) attention implemented as the custom layer TemporalAttention with weights W (128×128), b and u (128×1): $128 \cdot 128 + 128 + 128 = 16,640$ parameters, matching the table.

5.3 Training procedure and losses (cell [40]; config.py)

Setting	Value
Optimiser / learning rate	Adam, 1e-3; ReduceLROnPlateau ×0.5, patience 3 (min 1e-5)
Batch size / max epochs / early stopping	64 / 40 / patience 6 on validation-bearing loss, best weights restored (LOBO)
RUL head loss	Huber, $\delta = 0.1$, on capped RUL / 120
Stage head loss	sparse categorical cross-entropy with balanced class weights (compute_class_weight('balanced'))
Loss weights	RUL 1.0, stage 0.5 (fixed, not tuned)
Seeds	LOBO: 42, 43; final: 42, 43, 44; TensorFlow deterministic ops enabled
Final training	No validation bearing: epochs fixed from LOBO median best epoch (at least 3): proposed = 3 (cell [47])
Hyper-parameter search	None (deliberate, to avoid over-fitting six bearings) (cell [84])

$$L = L_{\text{Huber}}^{\text{RUL}} + 0.5 L_{\text{CE}}^{\text{stage}}$$

$$L_{\delta}(e) = \frac{1}{2}e^2 \text{ if } |e| \leq \delta; \quad L_{\delta}(e) = \delta(|e| - \frac{\delta}{2}) \text{ otherwise} \quad (\delta = 0.1)$$

5.4 How each head is used and evaluated

- **RUL head** → predicted capped RUL in minutes. Scored with MAE vs the *capped* true RUL, with detection AUC (score = -predicted RUL for “failure within H minutes”) and with the official PHM-2012 score.
- **Stage head** → probabilities of three stages (“native” stage). Because other models have no such head, the main comparison uses a **uniform rule**: stage derived from the *predicted RUL* with the same 20/60-min thresholds for every model. Native stage-head results are reported separately for the proposed model, the dual-head ablation and Random Forest (cell [71]).
- **Alarm** (identical for all models): predicted stage ≥ Warning for 3 consecutive windows (ALARM_PERSISTENCE=3).

Cell [91] item 8–11 gives the *design arguments* (CNN: local patterns; BiLSTM: trend; attention: inspectable weights; dual head: alarm level and possible regularisation). These are **arguments, not measured advantages**. The measured result (Section 6.6): attention was **MIXED** (2 improvements, 0 degradations, 6 no difference) and the dual head was **MIXED** (1 / 1 / 6) in cell [86].

6. Results and graphs

Every figure below is an **existing saved figure** from `final_project/figures/` (identical to the notebook's embedded output; shown here as down-sized copies). Each has the same five-part explanation. "Source cell" is the zero-based notebook cell index.

6.1 Dataset, vibration signals and degradation patterns

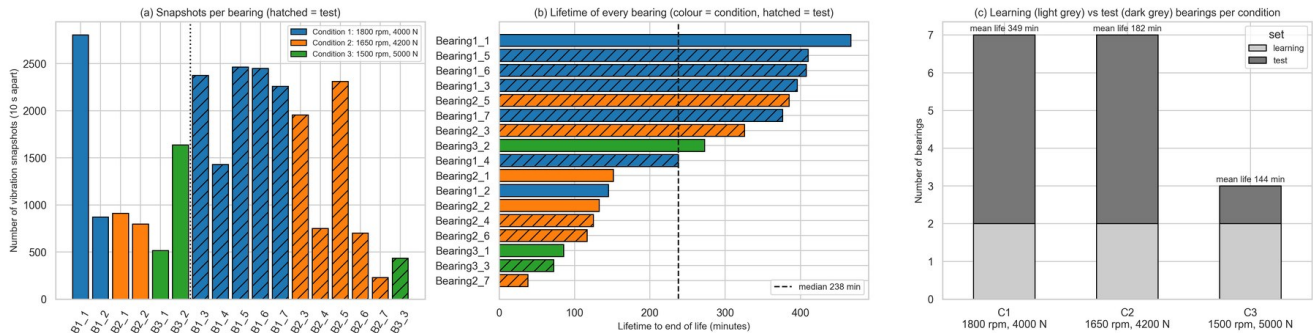


Figure G2. Dataset overview: snapshots, lifetimes and operating conditions. (Source: notebook cell [8]; saved file `final_project/figures/fig01_dataset_overview.png`)

What is shown	(a) number of 10-second snapshots per bearing (hatched = test); (b) lifetime of every bearing sorted by length, colour = operating condition; (c) learning versus test bearings per condition with mean life.
How to read it	Panel (b): longer bar = longer life. The dashed line is the median lifetime. Panel (c): light grey = learning, dark grey = test.
What the result shows	Lifetimes range from 38 to 467 minutes (factor ≈ 12) ; per condition (min/median/max) C1 145.0/395.7/467.0, C2 38.2/132.7/385.0, C3 72.2/85.7/272.7 min. Condition 3 has only 3 bearings in total (2 learning).
Reasonable conclusion	Bearings of the same type live for very different times, and learning data is small. This is the difficulty named by the challenge organisers (cell [9]).
Do NOT conclude	Nothing about model accuracy. Bearing1_4 shows 237.8 min here because all recorded files are counted; it is cut to 195.2 min at its official end of life later.

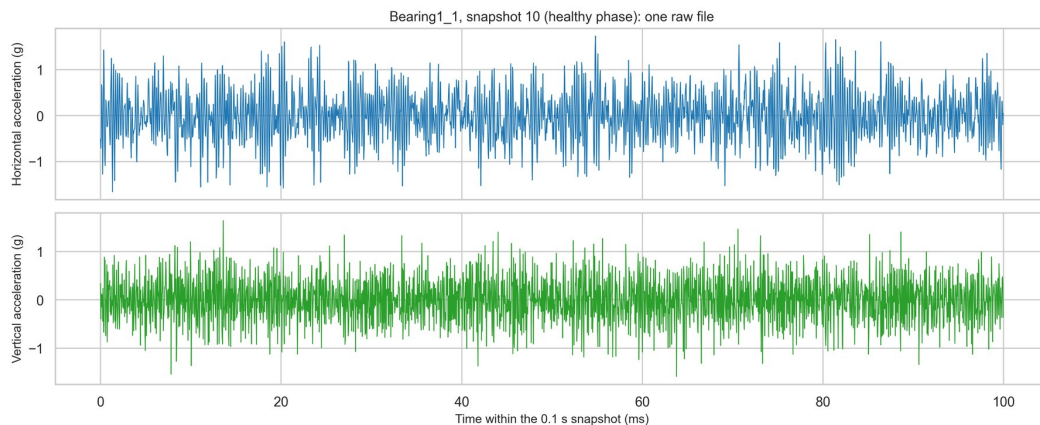


Figure G3. One raw 0.1 s snapshot of Bearing1_1 (healthy phase, snapshot 10). (Source: notebook cell [11]; saved file `final_project/figures/fig02_sample_raw_signal.png`)

What is shown	The two raw accelerometer channels (horizontal and vertical) over the 100 ms of one file (2560 samples).
How to read it	x-axis: time inside the snapshot (ms); y-axis: acceleration in g.
What the result shows	Horizontal peak 1.74 g, vertical peak 1.64 g; a low-level, noise-like waveform. The time-base check (39 μ s between samples = 25.6 kHz) passed.
Reasonable conclusion	This is what raw data look like; it justifies units (g), channels and sampling rate.
Do NOT conclude	That this single healthy file represents the whole bearing life.

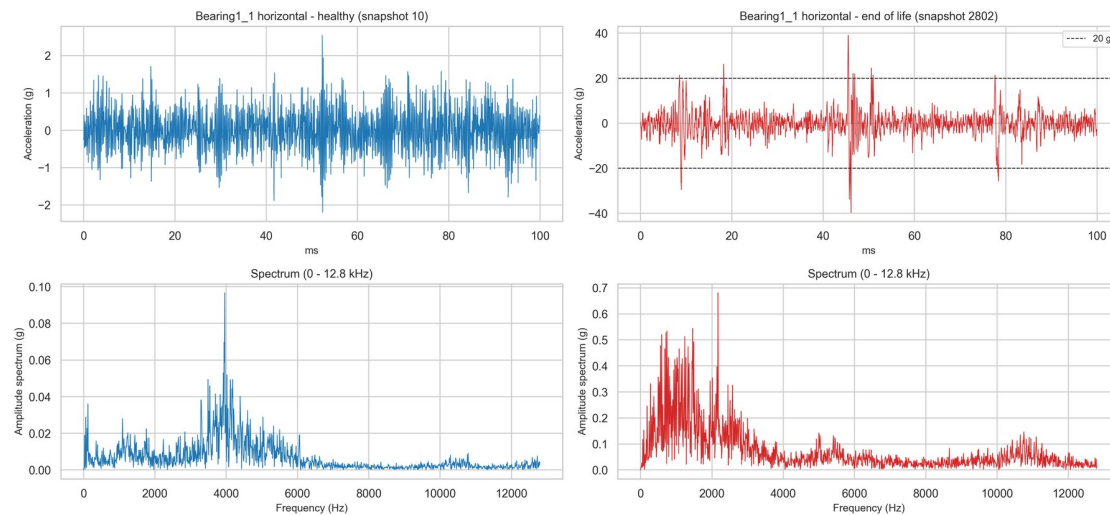


Figure G4. Healthy versus end-of-life snapshot of Bearing1_1: waveform (top) and amplitude spectrum (bottom). (Source: notebook cell [12]; saved file final_project/figures/fig03_healthy_vs_degraded.png)

What is shown	Left column: snapshot 10 (healthy). Right column: the last snapshot (2802). Top: horizontal acceleration with ± 20 g lines on the right. Bottom: FFT amplitude spectrum up to 12.8 kHz.
How to read it	Compare the y-axis scales: the healthy waveform is within $\approx \pm 2$ g, the end-of-life one reaches the 20 g region. In the spectrum look at how broadly the energy is spread.
What the result shows	Impacts far above healthy level and energy spread over a broad frequency range at end of life.
Reasonable conclusion	Motivates amplitude features (RMS, peak, kurtosis) and spectral-shape/band features (centroid, spread, band energies) (cell [12] text).
Do NOT conclude	That every bearing fails in this way, or that the model has been shown anything here: it is one bearing and two snapshots.

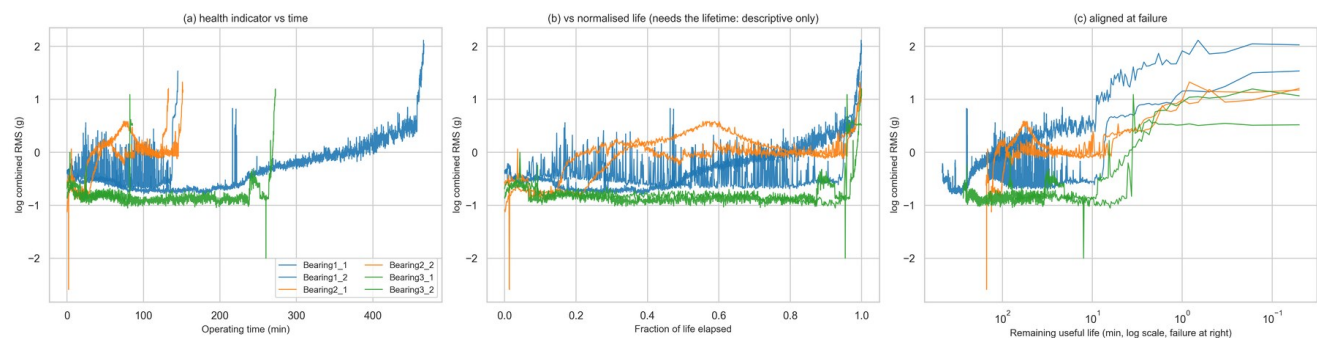


Figure G5. Health indicator (log of combined RMS) of the six learning bearings. (Source: notebook cell [14]; saved file final_project/figures/fig04_health_indicator_learning.png)

What is shown	(a) indicator versus operating time; (b) versus fraction of life elapsed (needs the lifetime, descriptive only); (c) all bearings aligned at failure on a log axis of remaining life.
How to read it	Colours = operating condition. In (c), failure is at the right; a curve that stays flat until the right edge means almost no warning in the vibration.
What the result shows	Bearing1_1 rises slowly over hundreds of minutes; Bearing3_1, 3_2 and 1_2 stay flat until the last minutes (final-phase length 3.8–9 min in the table of cell [14]); Bearing2_1 and 2_2 show early level changes that are not clearly damage (cell [15]).
Reasonable conclusion	Real bearings degrade in different ways ; a model cannot warn early if the vibration looks healthy until minutes before failure. This is a property of the data (cell [15]).
Do NOT conclude	That the final-phase table is an achievable warning time: it is hindsight-based, uses learning bearings only, and was never used as a label.

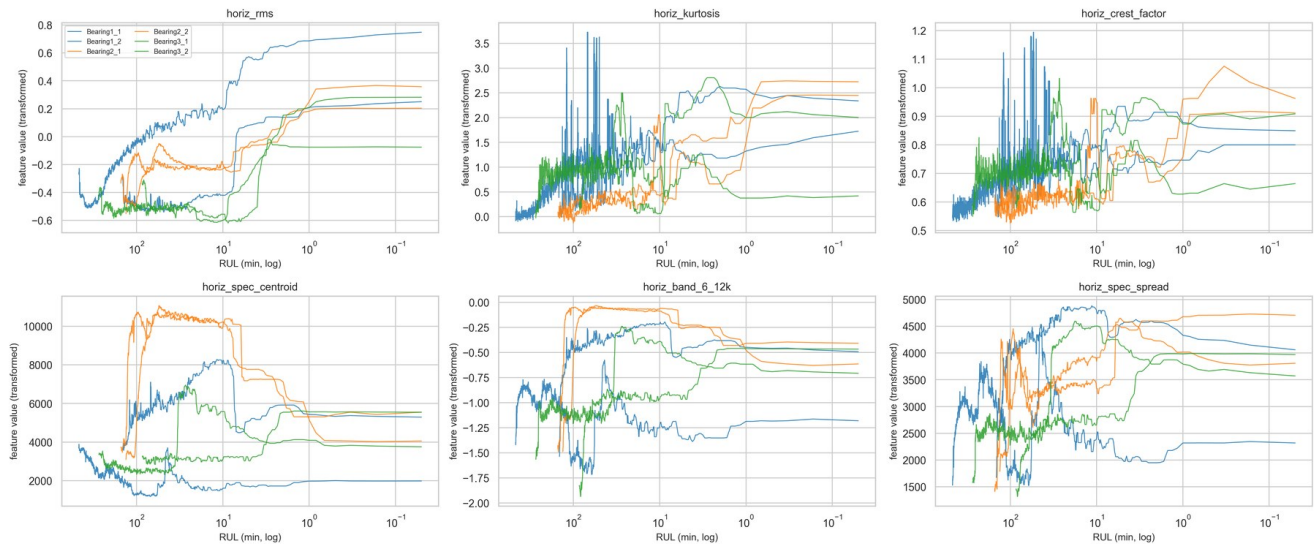


Figure G6. Six selected features against remaining life on the learning bearings. (Source: notebook cell [17]; saved file final_project/figures/fig05_feature_vs_life_learning.png)

What is shown	Median-smoothed (for readability only) feature values versus remaining life (log axis, failure at right), one line per learning bearing.
How to read it	Look for where each line starts to move and whether lines of different bearings move the same way.
What the result shows	Features move mostly within the last tens of minutes, and their absolute levels differ between bearings (cell [17] text). Only 2 of 34 features have a consistent whole-life trend (cell [19]).
Reasonable conclusion	No single feature or fixed threshold is a robust health indicator across bearings; hence a learned model over a window of many features, scaled with training-bearing statistics.
Do NOT conclude	That the feature screening chose the model inputs: it did not (cell [16]); all 34 features are always used.

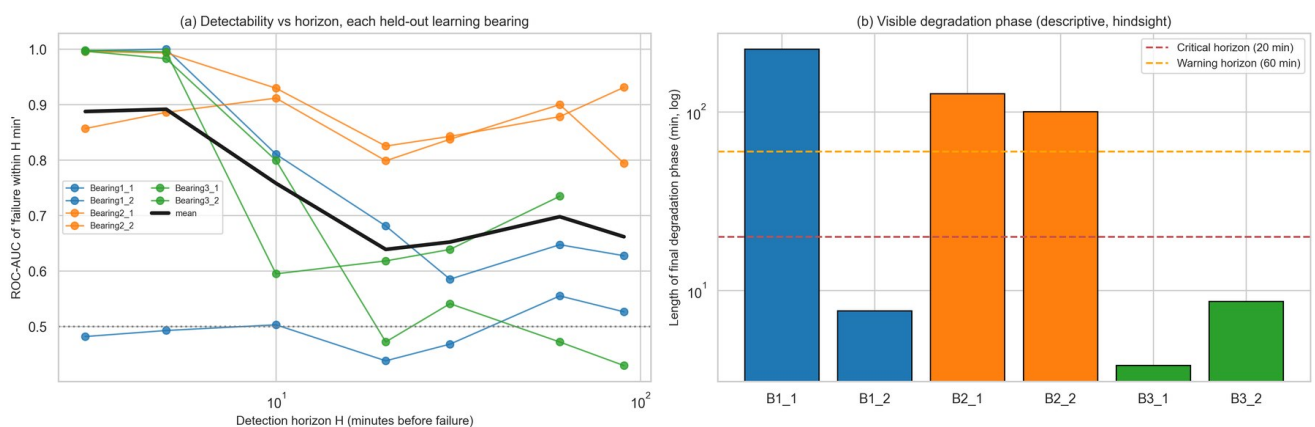


Figure G7. How well can vibration detect “failure within H minutes”? (Random-Forest probe, learning bearings, LOBO). (Source: notebook cell [27]; saved file final_project/figures/fig08_horizon_detectability.png)

What is shown	(a) ROC-AUC of detecting failure within H minutes for each held-out learning bearing and their mean (black), for $H = 3 \dots 90$ min. (b) Length of the final visible degradation phase of each learning bearing (log axis) against the Critical (20 min) and Warning (60 min) horizons.
How to read it	AUC 0.5 = chance, 1.0 = perfect. Higher is better. In (b) a bar below a dashed line means the visible degradation is shorter than that horizon.
What the result shows	Mean AUC falls from 0.89 at 5 min to 0.64 at 20 min and 0.70 at 60 min (cell [29]); some bearings are near chance at long horizons (e.g. Bearing1_2 ≈ 0.5 up to 10 min). Several bearings' visible phase is shorter than 60 min.
Reasonable conclusion	Detectability depends strongly on the horizon and on the bearing; this is why every later model is evaluated at several horizons.
Do NOT conclude	That this holds for the proposed model: the probe is a Random Forest on learning bearings. Do not read it as a test-set result.

6.2 Train / validation / test protocol

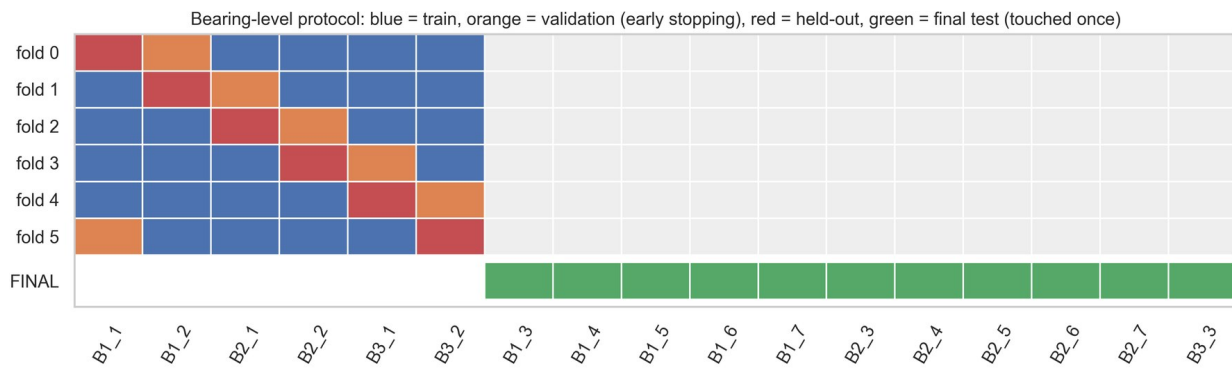


Figure G8. Bearing-level protocol (6 LOBO folds + final test). (Source: notebook cell [35]; saved file `final_project/figures/fig09_split_protocol.png`)

What is shown	Rows are LOBO folds and the FINAL row; columns are the 17 bearings. Blue = training, orange = validation (early stopping only), red = held-out (scored), green = final test bearings (touched once).
How to read it	Each LOBO row has exactly one red and one orange cell; grey cells are unused in that fold.
What the result shows	Test bearings (green, columns 7–17) never appear in any fold. Every learning bearing is held out exactly once.
Reasonable conclusion	The evaluation unit is the bearing, and the test set is separate from model development.
Do NOT conclude	That the folds are statistically independent of one another in training data: each pair of folds shares 3 of 4 training bearings.

Same model, same bearings, two protocols (output of cell [35]):

Protocol	MAE (min)	AUC (failure within 60 min)
RANDOM window split (wrong)	1.466	0.998
Bearing-level LOBO (correct)	42.197	0.698

This table is the single most convincing slide for “why split by bearing”. The only difference between the rows is leakage of near-duplicate windows.

6.3 Training and validation history

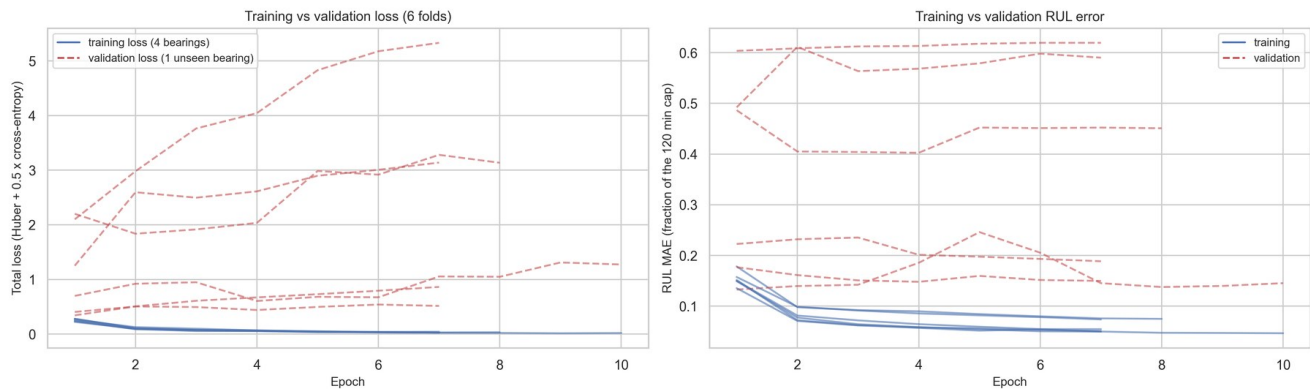


Figure G9. Training versus validation loss and RUL error of the proposed model, seed 42, all six LOBO folds. (Source: notebook cell [42]; saved file `final_project/figures/fig_training_curves.png`)

What is shown	Left: total loss (Huber + 0.5 × cross-entropy) per epoch; blue solid = training on 4 bearings, red dashed = validation on 1 unseen bearing. Right: RUL MAE as a fraction of the 120-min cap.
How to read it	A healthy training run has both curves falling together. A growing red curve while blue falls means over-fitting.
What the result shows	Training loss falls below 0.1 by epoch 2 and keeps shrinking; validation loss stays high or rises in most folds (up to ≈ 5). Median best epoch = 1.0. Stage accuracy on training bearings exceeds that on the unseen validation bearing by 49 percentage points at the end of training (cell [42]).
Reasonable conclusion	The network memorises the 4 training bearings almost immediately; capacity is not the bottleneck, data is (cell [43]). Early stopping and a short fixed final training (3 epochs) were necessary.
Do NOT conclude	That the architecture is wrong or that nothing was learned: validation curves say nothing yet about whether the architecture helps; the model comparison (Section 6.4) answers that.

6.4 Validation (LOBO) comparison of all models

Mean ± standard deviation over the 6 held-out bearings (the bearing is the independent unit). Source: `artifacts/lobo_model_comparison.csv`, printed by cell [45]. For RUL errors lower is better; for AUC and macro-F1 higher is better.

Model	MAE all (min)	MAE actionable (min)	AUC@10	AUC mean-H	macro-F1	false-alarm rate
Constant	40.310 ± 4.179	53.373 ± 11.107	0.500 ± 0.000	0.500 ± 0.000	0.246 ± 0.058	0.000 ± 0.000
Ridge	55.531 ± 23.253	31.076 ± 18.026	0.603 ± 0.174	0.615 ± 0.198	0.267 ± 0.214	0.535 ± 0.427
Random Forest	44.125 ± 17.331	38.902 ± 21.387	0.776 ± 0.162	0.717 ± 0.159	0.342 ± 0.165	0.365 ± 0.372
Gradient Boosting	43.007 ± 16.689	39.680 ± 20.811	0.781 ± 0.161	0.725 ± 0.091	0.335 ± 0.141	0.362 ± 0.363
CNN	46.152 ± 19.683	28.946 ± 21.476	0.803 ± 0.183	0.803 ± 0.172	0.294 ± 0.163	0.563 ± 0.356
LSTM	42.477 ± 21.791	27.978 ± 26.595	0.940 ± 0.049	0.886 ± 0.104	0.373 ± 0.160	0.454 ± 0.341
CNN + LSTM	43.131 ± 18.977	24.150 ± 16.352	0.883 ± 0.106	0.852 ± 0.128	0.325 ± 0.182	0.517 ± 0.339
CNN + BiLSTM	46.738 ± 20.546	28.407 ± 22.156	0.893 ± 0.077	0.873 ± 0.116	0.314 ± 0.179	0.537 ± 0.346
CNN+BiLSTM+Attn (single)	44.922 ± 18.441	23.340 ± 17.162	0.848 ± 0.144	0.830 ± 0.149	0.315 ± 0.156	0.586 ± 0.345

Model	MAE all (min)	MAE actionable (min)	AUC@10	AUC mean-H	macro-F1	false-alarm rate
CNN+BiLSTM (dual, no attn)	47.172 ± 24.217	31.326 ± 25.701	0.879 ± 0.107	0.827 ± 0.126	0.317 ± 0.197	0.506 ± 0.339
PROPOSED CNN+BiLSTM+Attn (dual)	45.510 ± 21.669	28.752 ± 25.005	0.870 ± 0.133	0.847 ± 0.118	0.351 ± 0.183	0.523 ± 0.324

Reading the table. (1) No model beats the constant predictor on overall MAE (constant 40.3 min; cell [48]), because most windows are healthy and a constant is hard to beat there. (2) All 10 non-constant models beat the constant on actionable-region MAE (constant 53.4 min). (3) Best detection AUC (mean over horizons) is **LSTM 0.886**; proposed 0.847; best tabular 0.725 (Gradient Boosting). (4) The standard deviations over bearings (0.09–0.20 for AUC) are large, so many differences are not meaningful. (5) The proposed model was **not** the model selected by the LOBO rules: selection.json picked LSTM (detection) and CNN+BiLSTM+Attention single head (actionable RUL); the proposed model is listed as “interpretable by design”.

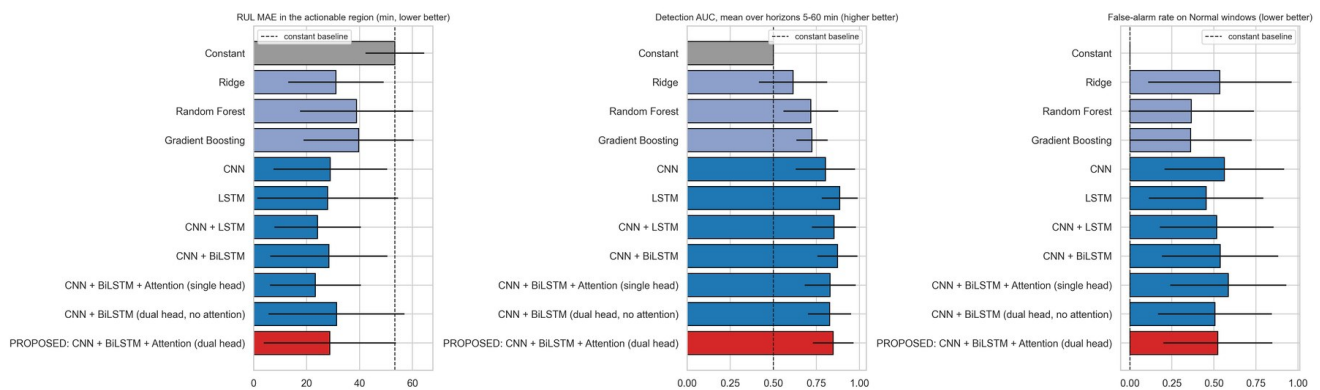


Figure G10. LOBO comparison bars: actionable RUL MAE, detection AUC and false-alarm rate. (Source: notebook cell [46]; saved file final_project/figures/fig_model_comparison_lobo.png)

What is shown	For every model, the mean over six held-out bearings (bar) and standard deviation across bearings (whiskers); dashed line = constant predictor; red = proposed.
How to read it	Left panel: shorter is better. Middle: longer is better. Right: shorter is better (but the constant has 0 only because it never alarms).
What the result shows	Neural sequence models tend to have higher AUC than tabular models; whiskers overlap heavily; false-alarm rates are 0.36–0.59 for all non-constant models.
Reasonable conclusion	Sequence models are promising for detecting imminent failure on the six learning bearings (H1 holds on LOBO: mean AUC difference +0.162, 95 % CI +0.092 ... +0.238, cell [86]).
Do NOT conclude	That the proposed model is the best: LSTM is higher on AUC and the differences within the whiskers are not significant with n = 6.

6.5 Final test on 11 unseen bearings

Source: artifacts/final_model_comparison_test.csv, printed by cell [69]; mean ± std over 3 training seeds (Constant and Ridge: one seed). Population: all windows of the 11 test bearings (16,934 windows), each scored once after the design was frozen.

Model	MAE all (min)	MAE actionable (min)	AUC@10	AUC mean-H	macro-F1	false alarm	PHM-2012 score
Constant	35.623	56.354	0.500	0.500	0.291	0.000	0.124
Ridge	39.066	45.216	0.664	0.640	0.422	0.198	0.040
Random Forest	32.640 ± 0.101	32.713 ± 0.124	0.654 ± 0.002	0.696 ± 0.002	0.427 ± 0.001	0.296 ± 0.001	0.078 ± 0.005
Gradient Boosting	30.146 ± 0.000	32.651 ± 0.000	0.694 ± 0.000	0.725 ± 0.000	0.471 ± 0.000	0.237 ± 0.000	0.037 ± 0.000
CNN	27.806 ± 0.889	42.575 ± 2.302	0.751 ± 0.030	0.732 ± 0.032	0.454 ± 0.019	0.146 ± 0.011	0.181 ± 0.022

Model	MAE all (min)	MAE actionable (min)	AUC@10	AUC mean-H	macro-F1	false alarm	PHM-2012 score
LSTM	32.324 ± 0.773	36.779 ± 2.906	0.739 ± 0.005	0.728 ± 0.011	0.450 ± 0.027	0.208 ± 0.022	0.143 ± 0.023
CNN + LSTM	30.610 ± 3.932	42.055 ± 1.535	0.719 ± 0.010	0.715 ± 0.009	0.423 ± 0.065	0.182 ± 0.059	0.221 ± 0.027
CNN + BiLSTM	34.205 ± 3.342	40.092 ± 7.291	0.737 ± 0.061	0.725 ± 0.055	0.410 ± 0.041	0.225 ± 0.070	0.106 ± 0.026
CNN+BiLSTM+Attn (single)	32.336 ± 5.166	41.184 ± 4.669	0.726 ± 0.041	0.715 ± 0.034	0.433 ± 0.050	0.202 ± 0.071	0.179 ± 0.023
CNN+BiLSTM (dual, no attn)	35.047 ± 3.243	34.110 ± 5.958	0.741 ± 0.029	0.748 ± 0.026	0.399 ± 0.045	0.277 ± 0.055	0.093 ± 0.031
PROPOSED CNN+BiLSTM+Attn (dual)	29.592 ± 5.308	38.072 ± 6.187	0.759 ± 0.024	0.749 ± 0.017	0.468 ± 0.045	0.172 ± 0.109	0.146 ± 0.088

What the test table says (cell [76], [86]). Detection AUC (mean over horizons): proposed **0.749**, constant 0.50; the proposed model has the highest mean but is **statistically tied** (within 0.020 / seed noise) with CNN and CNN+BiLSTM (dual, no attention). Actionable-region MAE: proposed 38.1 min vs constant 56.4 min, but the lowest is **Gradient Boosting 32.7**, tied with Random Forest and the dual-head no-attention model; the proposed model is **not** in that tie group. Overall MAE: proposed 29.6 vs constant 35.6 min. False-alarm rate (uniform rule): proposed 0.17 (RF 0.30, GB 0.24). The ranking of models on LOBO is **not reproduced** on test (rank correlation +0.47 for detection, -0.52 for actionable RUL error).

Seed-to-seed variability of the proposed model (final_metrics_per_model_seed.csv; the saved model file and the per-bearing figures use seed 42):

Seed	MAE all	MAE actionable	RMSE	AUC@10	AUC mean-H	macro-F1	false alarm	PHM-2012
42	23.83	44.43	38.74	0.759	0.742	0.521	0.053	0.097
43	30.65	37.72	45.36	0.782	0.768	0.444	0.197	0.248
44	34.29	32.07	48.24	0.735	0.736	0.441	0.267	0.094

Actionable MAE varies from 32.1 to 44.4 min and the false-alarm rate from 0.05 to 0.27 across seeds with identical data and settings. RMSE is computed by the code and stored in this CSV, but it is not shown in any notebook cell output; treat it as an available-but-unreviewed figure.

Official PHM-2012 single-inspection score (cell [69])

Bearing	Actual RUL (s)	Predicted RUL (s)	% error	A _i
Bearing1_3	5730	6376.9	-11.3	0.209
Bearing1_4	339	53.1	84.3	0.054
Bearing1_5	1610	6873.7	-326.9	0.000
Bearing1_6	1460	2827.1	-93.6	0.000
Bearing1_7	7570	468.4	93.8	0.039
Bearing2_3	7530	6932.8	7.9	0.760
Bearing2_4	1390	7102.0	-410.9	0.000
Bearing2_5	3090	5647.6	-82.8	0.000
Bearing2_6	1290	2734.1	-111.9	0.000
Bearing2_7	580	7157.0	-1134.0	0.000
Bearing3_3	820	1303.0	-58.9	0.000

Proposed, seed 42: **0.097**. Seed mean over 3 seeds: **0.146 ± 0.088**. Constant predictor: **0.124**. Random Forest 0.078. Higher is better; 1 is perfect. Negative % error means the model predicted *more* life than remains (late warning), which the score punishes hardest.

Interpretation. The score is dominated by a few bearings (in seed 42, Bearing2_3 alone contributes 0.760 of a total of about 1.06). At the truncation points of the abrupt bearings (1_5, 2_4, 2_5, 2_6, 2_7, 3_3) the model still predicts a healthy bearing, giving $A_i = 0$. Seed 42 scores *below* the constant predictor and the 3-seed mean is only slightly above it with a std of 0.088, so **do not claim the model beats the constant on the official score**. The notebook itself says the score is “very noisy” and “not comparable with challenge leaderboards” because the predictor is capped at 120 min = 7200 s (Bearings 1_7 and 2_3 have official RUL above 7200 s) (cells [69], [84]).

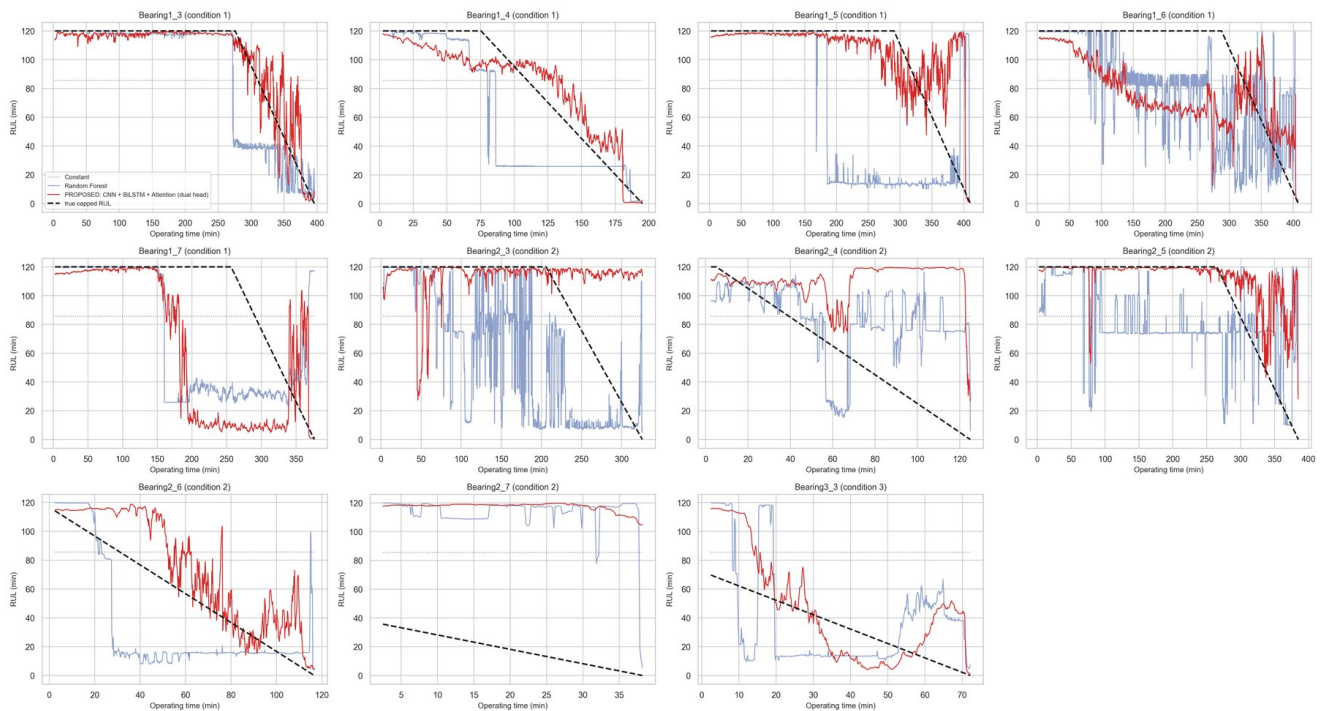


Figure G11. Predicted versus true capped RUL over the whole life of every test bearing (seed 42). (Source: notebook cell [70]; saved file final_project/figures/fig_per_bearing_test.png)

What is shown	One panel per test bearing. Black dashed = true capped RUL; red = proposed model; blue = Random Forest; grey dotted = constant.
How to read it	A model with skill would follow the black dashed line downward ahead of failure (right-hand end). Lines far above the dashed line near the end mean the model predicts too much remaining life (late/no warning).
What the result shows	Bearing1_3, 2_5 and 2_6 follow the decline only noisily and mostly in the last part of life; Bearing1_4 declines smoothly (somewhat too early in the first half); Bearing2_3 and 2_7 stay near 115–120 min almost to the end (no warning); Bearing1_7 drops to ≈ 10 min at ≈ 160 – 190 min and stays there for roughly 150 min although the true capped RUL is still 120 (a long false alarm); Bearing2_4 is predicted healthy (≈ 120 min) for roughly its last 55 minutes.
Reasonable conclusion	Performance is very uneven across bearings: reasonable on gradual bearings, poor on abrupt ones (cell [70] text: this is the direct visual answer to “does it warn in time?”).
Do NOT conclude	That the model gives accurate minute-level RUL (it does not) or that a single seed represents all seeds.

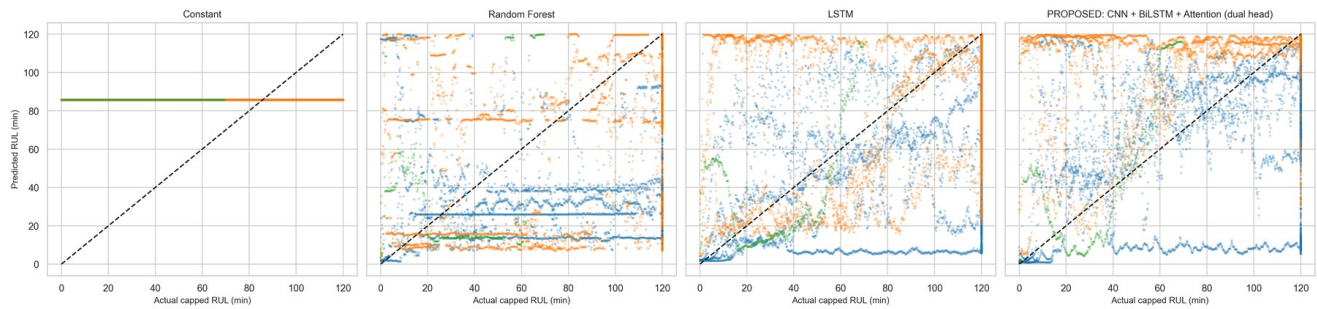


Figure G12. Actual versus predicted capped RUL, all test windows, four models (seed 42). (Source: notebook cell [70]; saved file final_project/figures/fig_actual_vs_predicted_test.png)

What is shown	Each dot is one test window: x = true capped RUL, y = predicted RUL; colour = operating condition; diagonal = perfect prediction.
How to read it	Dots near the dashed diagonal are good. A horizontal cloud means the model ignores the input (the constant predictor is exactly horizontal).
What the result shows	The proposed model and others scatter widely around the diagonal; most mass is at true RUL = 120 (healthy windows).
Reasonable conclusion	There is signal, but it is far from exact RUL regression.
Do NOT conclude	That a low overall MAE means good RUL estimates: the overall MAE is dominated by healthy windows.

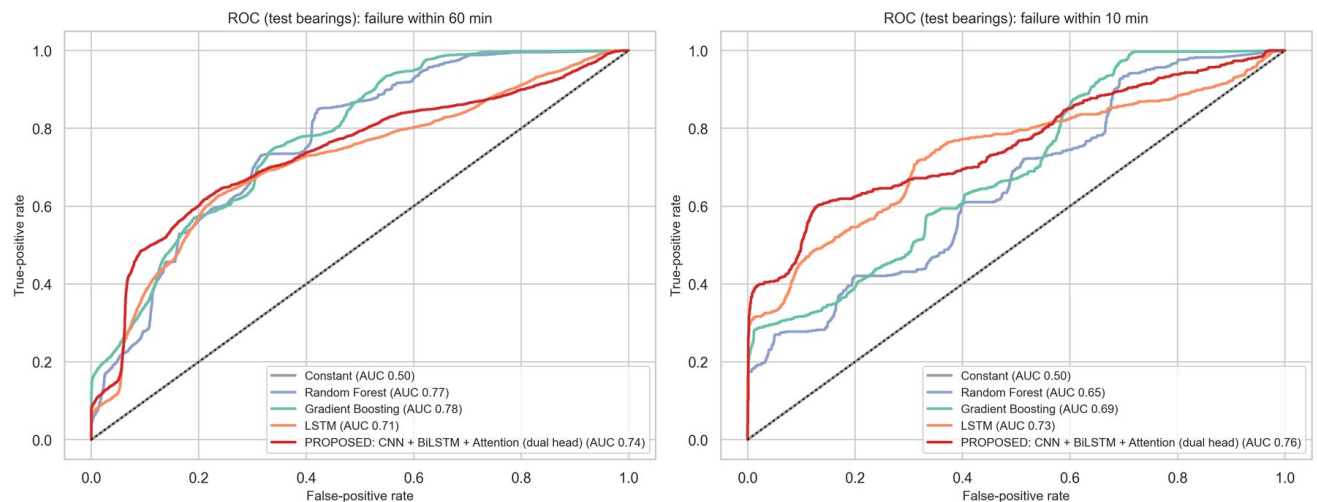


Figure G13. ROC curves for detecting “failure within 60 min” and “within 10 min” on the test bearings (pooled, seed 42). (Source: notebook cell [70]; saved file final_project/figures/fig_roc_test.png)

What is shown	True-positive rate versus false-positive rate when the alarm threshold on predicted RUL is varied; legend gives the pooled AUC.
How to read it	A curve further to the top-left is better; the diagonal is chance. Cell [55]/[70] text: the longer horizon (60 min, left) is much harder than the short one (10 min, right).
What the result shows	Curves lie above the diagonal for the learned models and equal the diagonal for the constant.
Reasonable conclusion	Imminent failure is recognisable better than chance when windows from all test bearings are pooled.
Do NOT conclude	That every bearing is detected well: pooled ROC hides per-bearing failures (see the per-bearing table below, e.g. Bearing2_4 AUC@60 = 0.10).

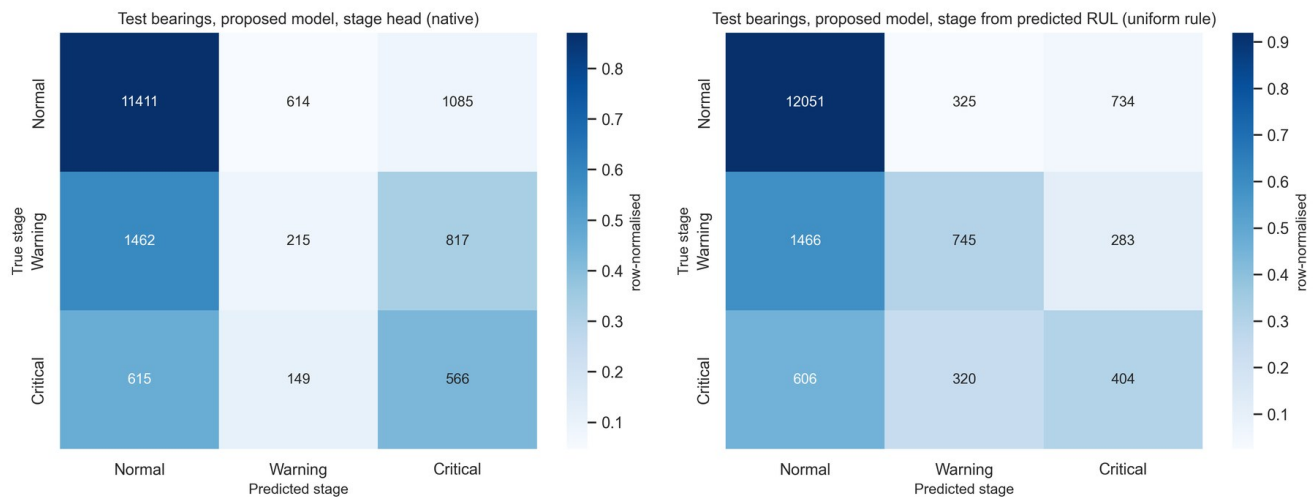


Figure G14. Confusion matrices of the proposed model on the test bearings (seed 42). (Source: notebook cell [70]; saved file final_project/figures/fig_confusion_test.png)

What is shown	Left: the model's own stage head. Right: stage derived from the predicted RUL by the uniform rule. Rows = true stage, columns = predicted stage; numbers = window counts, colour = fraction of the row.
How to read it	The diagonal is correct classification. Read each row as "of the windows that truly belong to this stage, how many were assigned to each stage?".
What the result shows	Normal is well recognised (11,411 / 13,110 native; 12,051 / 13,110 uniform). Warning is poorly recognised: native 215 of 2,494 correct; uniform 745. Critical: native 566 of 1,330 correct; uniform 404.
Reasonable conclusion	The stage head is biased toward Normal/Critical and misses most Warning windows; stage-from-RUL is better on Warning, worse on Critical (cell [71]: native recall 0.70 / 0.10 / 0.55; uniform rule 0.76 / 0.33 / 0.39).
Do NOT conclude	That accuracy (0.60 native, 0.67 uniform) shows good performance: it is dominated by the large Normal class. Use per-class recall and macro-F1.

Per-bearing results of the proposed model (seed 42, cell [72]; artifact final_per_bearing_proposed.csv)

Bearing	Windows	MAE all	MAE act.	AUC@10	AUC@60	macro-F1	False alarm	Alarm at end	Sustained lead (min)
Bearing1_3	2360	5.95	17.81	0.99	0.99	0.84	0.01	yes	19.67
Bearing1_4	1157	13.36	13.91	0.99	1.00	0.83	0.00	yes	41.32
Bearing1_5	2448	14.59	58.92	0.91	0.74	0.50	0.00	yes	7.67
Bearing1_6	2433	33.58	23.08	0.96	0.86	0.58	0.09	yes	4.17
Bearing1_7	2244	42.20	35.39	0.90	0.73	0.39	0.41	yes	8.00
Bearing2_3	1940	25.39	85.36	0.65	0.70	0.29	0.03	no	n/a
Bearing2_4	736	49.72	84.08	0.44	0.10	0.23	0.00	yes	1.17
Bearing2_5	2296	13.94	58.77	0.84	0.96	0.36	0.00	yes	0.33
Bearing2_6	686	19.54	16.11	0.88	0.99	0.63	0.00	yes	6.50
Bearing2_7	215	99.55	99.55	0.88	n/a	0.00	n/a	no	n/a
Bearing3_3	419	22.29	17.95	0.50	1.00	0.47	0.00	yes	44.17

"Alarm at end" = the alarm is still on at the last window before failure; "sustained lead" = remaining life when that final uninterrupted alarm run started (n/a if no alarm at the end). n/a for AUC@60 or false alarm means the quantity is undefined for that bearing (e.g. Bearing2_7 lives 38 min, so no window has RUL > 60 min).

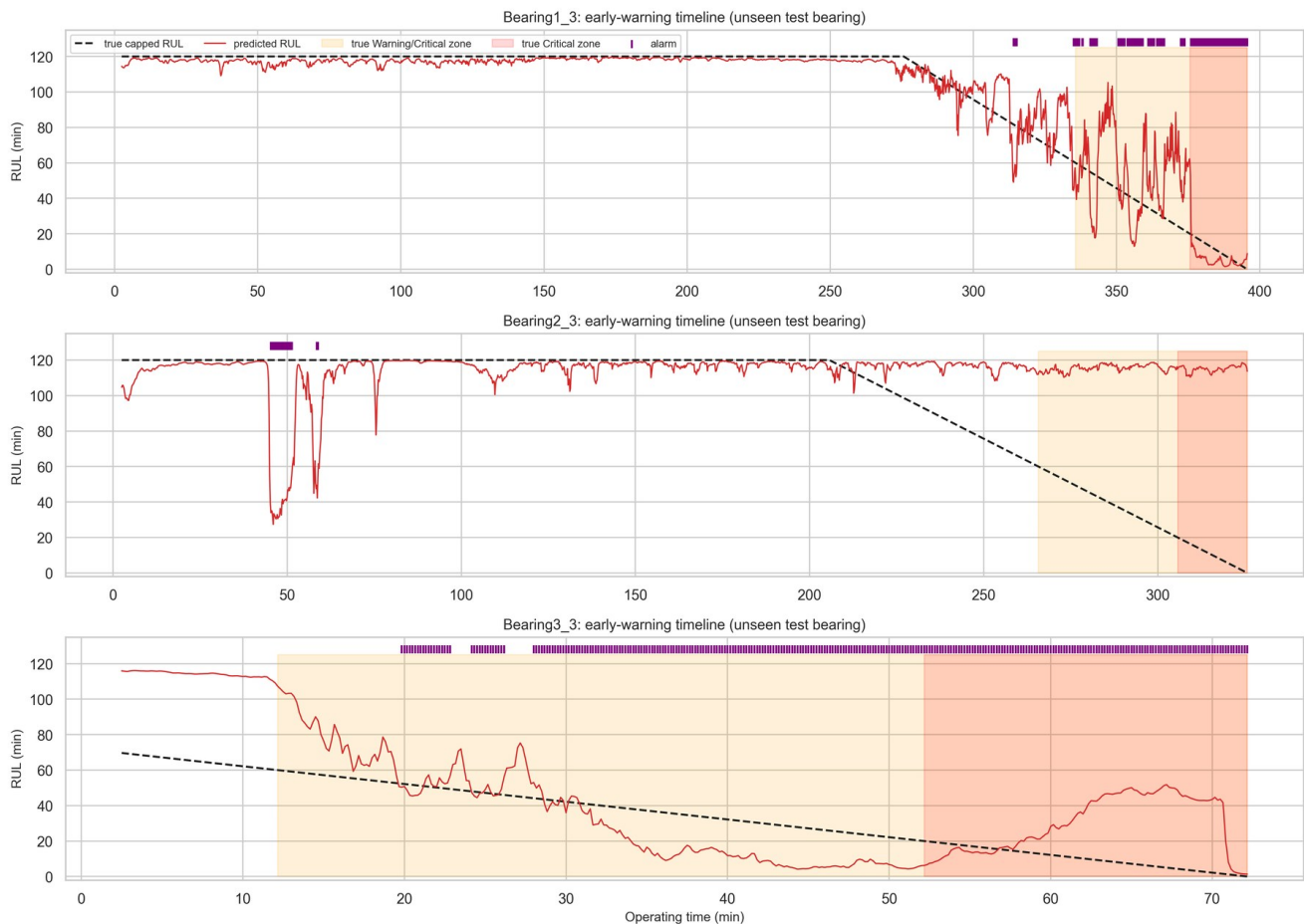


Figure G15. Early-warning timelines for Bearing1_3, 2_3 and 3_3 (the first test bearing of each condition, chosen by a rule fixed in advance). (Source: notebook cell [72]; saved file final_project/figures/fig_early_warning_test.png)

What is shown	True capped RUL (black dashed), predicted RUL (red), true Warning/Critical zones (shaded), and alarms (purple ticks) under the uniform rule (3 consecutive windows predicted $\geq$ Warning).
How to read it	An alarm tick inside the shaded zone is a correct warning; a tick in the white area is a false alarm; a shaded zone without ticks is a missed warning.
What the result shows	Bearing1_3: isolated alarms from ≈ 315 min (true RUL ≈ 80), intermittent in the Warning zone, continuous from ≈ 376 min (Critical zone). Bearing2_3: the model predicts ≈ 118 min throughout the Warning/Critical zones (no alarm, abrupt bearing); there are early false alarms near 45–60 min. Bearing3_3: alarm is on from ≈ 20 min of its 72-minute life, i.e. a sustained lead of ≈ 44 min.
Reasonable conclusion	Useful warnings exist for some bearings (1_3, 1_4, 3_3 in seed 42), and are missing or late for others.
Do NOT conclude	That early warning is demonstrated for all bearings: H3 is not supported (3/11 useful warnings).

6.6 Baselines and ablation study (paired bootstrap, 95 % CI)

Baselines **do exist** in this notebook (constant, Ridge, Random Forest, Gradient Boosting, LSTM, CNN, CNN+LSTM, CNN+BiLSTM). The ablation compares one change at a time with a bootstrap 95 % confidence interval over the paired bearings (6 LOBO folds; 11 test bearings). A component is called supported only if the interval excludes 0 in the helpful direction. Multiple-seed experiments also exist (2 seeds LOBO, 3 seeds final). Source: ablation_lobo.csv, ablation_test.csv; cells [51], [73].

Step (test bearings, n = 11)	Metric	Mean diff	95 % CI	Units improved	Verdict
CNN -> CNN+LSTM (add recurrence)	detection AUC at 10 min	-0.055	-0.110 ... -0.002	3/11	measurable degradation
CNN -> CNN+LSTM (add recurrence)	actionable RUL MAE (min)	-0.396	-6.187 ... +3.527	4/11	no measurable difference (CI includes 0)

Step (test bearings, n = 11)	Metric	Mean diff	95 % CI	Units improved	Verdict
LSTM -> BiLSTM (make it bidirectional)	detection AUC at 10 min	+0.020	-0.026 ... +0.074	6/11	no measurable difference (CI includes 0)
LSTM -> BiLSTM (make it bidirectional)	actionable RUL MAE (min)	-1.976	-7.567 ... +3.523	7/11	no measurable difference (CI includes 0)
add attention (single head)	detection AUC at 10 min	-0.024	-0.072 ... +0.022	4/11	no measurable difference (CI includes 0)
add attention (single head)	actionable RUL MAE (min)	+1.099	-4.000 ... +6.421	6/11	no measurable difference (CI includes 0)
add second head (dual head)	detection AUC at 10 min	+0.065	+0.002 ... +0.138	8/11	measurable improvement
add second head (dual head)	actionable RUL MAE (min)	-2.816	-11.652 ... +3.416	4/11	no measurable difference (CI includes 0)
add second head WITHOUT attention	detection AUC at 10 min	-0.009	-0.070 ... +0.054	5/11	no measurable difference (CI includes 0)
add second head WITHOUT attention	actionable RUL MAE (min)	-5.567	-14.464 ... +1.556	5/11	no measurable difference (CI includes 0)
add attention to the dual-head model	detection AUC at 10 min	+0.050	+0.012 ... +0.087	9/11	measurable improvement
add attention to the dual-head model	actionable RUL MAE (min)	+3.851	-2.758 ... +10.961	4/11	no measurable difference (CI includes 0)
add CNN in front of the LSTM	detection AUC at 10 min	-0.026	-0.080 ... +0.021	4/11	no measurable difference (CI includes 0)
add CNN in front of the LSTM	actionable RUL MAE (min)	+5.198	+2.707 ... +7.703	2/11	measurable degradation

Validation (LOBO) ablation – only the four steps about the proposed model’s two novel components (all 7 steps are in `ablation_lobo.csv`):

Step (LOBO, n = 6)	Metric	Mean diff	95 % CI	Units improved	Verdict
add attention (single head)	detection AUC (mean over horizons)	-0.042	-0.103 ... +0.003	2/6	no measurable difference (CI includes 0)
add attention (single head)	actionable RUL MAE (min)	-5.067	-9.621 ... -1.860	6/6	measurable improvement
add attention (single head)	stage macro-F1	+0.001	-0.021 ... +0.024	2/6	no measurable difference (CI includes 0)
add second head (dual head)	detection AUC (mean over horizons)	+0.017	-0.031 ... +0.071	4/6	no measurable difference (CI includes 0)
add second head (dual head)	actionable RUL MAE (min)	+5.412	+0.067 ... +12.647	1/6	measurable degradation
add second head (dual head)	stage macro-F1	+0.036	-0.003 ... +0.078	3/6	no measurable difference (CI includes 0)
add second head WITHOUT attention	detection AUC (mean over horizons)	-0.046	-0.152 ... +0.038	2/6	no measurable difference (CI includes 0)
add second head WITHOUT attention	actionable RUL MAE (min)	+2.918	-1.572 ... +7.657	3/6	no measurable difference (CI includes 0)
add second head WITHOUT attention	stage macro-F1	+0.003	-0.083 ... +0.063	4/6	no measurable difference (CI includes 0)
add attention to the dual-head model	detection AUC (mean over horizons)	+0.021	-0.048 ... +0.103	2/6	no measurable difference (CI includes 0)
add attention to the dual-head model	actionable RUL MAE (min)	-2.573	-6.990 ... +0.953	3/6	no measurable difference (CI includes 0)

Step (LOBO, n = 6)	Metric	Mean diff	95 % CI	Units improved	Verdict
add attention to the dual-head model	stage macro-F1	+0.034	-0.003 ... +0.087	3/6	no measurable difference (CI includes 0)

Combined verdict (cell [86], computed by code from every contrast on LOBO and test). Attention: **MIXED** – 2 improvements, 0 degradations, 6 no measurable difference. Dual head: **MIXED** – 1 improvement, 1 degradation, 6 no difference. “Mixed” means the effect changes sign or significance between metrics or between validation and test, so **no robust claim of improvement can be made**. Note: with 21 + 14 comparisons and no multiple-comparison correction, even the “measurable” entries should be treated as exploratory. “No measurable difference” is not evidence of harm; it means 6 or 11 bearings cannot distinguish the step from zero.

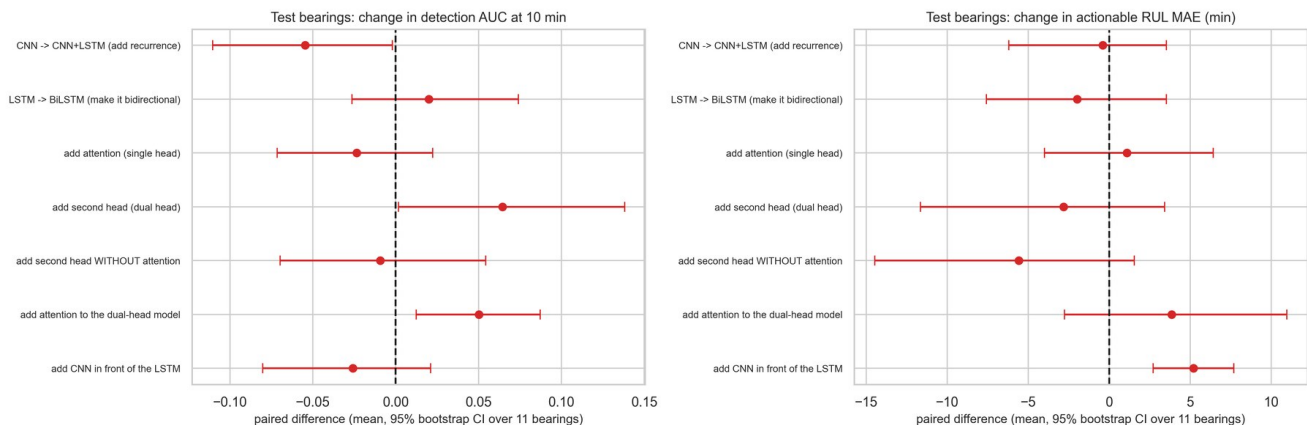


Figure G16. Test-bearing ablation: paired change caused by each architectural step. (Source: notebook cell [73]; saved file final_project/figures/fig_ablation_test.png)

What is shown	Each dot is the mean paired difference when a component is added; the bar is the 95 % bootstrap CI over the 11 test bearings. Left: detection AUC at 10 min; right: actionable RUL MAE.
How to read it	The vertical dashed line is zero. For AUC, right of zero = better; for MAE, left of zero = better. A bar that crosses zero = no measurable difference.
What the result shows	Adding the second head or attention to the dual-head model raises AUC@10 (CI above 0: +0.065 and +0.050); no step lowers actionable MAE measurably; adding the CNN in front of the LSTM <i>worsens</i> actionable MAE (+5.2 min).
Reasonable conclusion	On the test bearings, the proposed combination has a measurable AUC@10 advantage over its ablations, consistent with it being in the detection tie group.
Do NOT conclude	That the same holds on validation (it does not: attention improved MAE but dual head degraded it) or that RUL accuracy improves (no).

6.7 Explainability

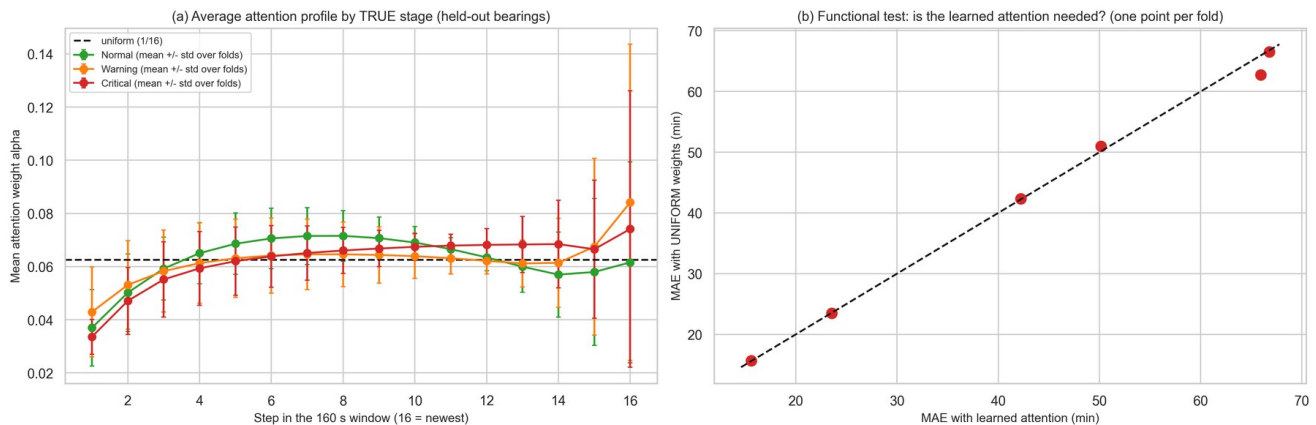


Figure G17. Attention profile by true stage and the “uniform-attention” functional test (held-out learning bearings). (Source: notebook cell [59]; saved file final_project/figures/fig_attention_lobo.png)

What is shown	(a) Average attention weight α at each of the 16 steps (16 = newest) for windows whose true stage is Normal, Warning or Critical, averaged over folds $\pm$ std; dashed = uniform 1/16. (b) Each dot is one fold: RUL MAE with the learned attention versus with attention replaced by uniform weights in the same trained network.
How to read it	(a) A flat line at the dashed level means the model attends equally to all steps. (b) Points on the diagonal mean replacing attention changes nothing.
What the result shows	Profiles are close to uniform (≈ 0.03 – 0.09 versus 0.0625 ; large fold-to-fold error bars); weight on the newest 4 steps is Normal 0.236 , Warning 0.274 , Critical 0.277 (uniform = 0.25), Critical > Normal in 5/6 folds. Replacing attention by uniform weights changes the prediction by 1.4 min on average (correlation 0.996) ; MAE $44.0 \rightarrow 43.6$ min.
Reasonable conclusion	The attention layer is functionally weak : it tells where the network put slightly more weight (a bit more on the newest steps near failure) but the prediction does not depend on it (cell [62]).
Do NOT conclude	That attention <i>explains why</i> the bearing is failing (causal), or that the architecture benefits from attention: attention weights are not explanations (Jain & Wallace [9]).

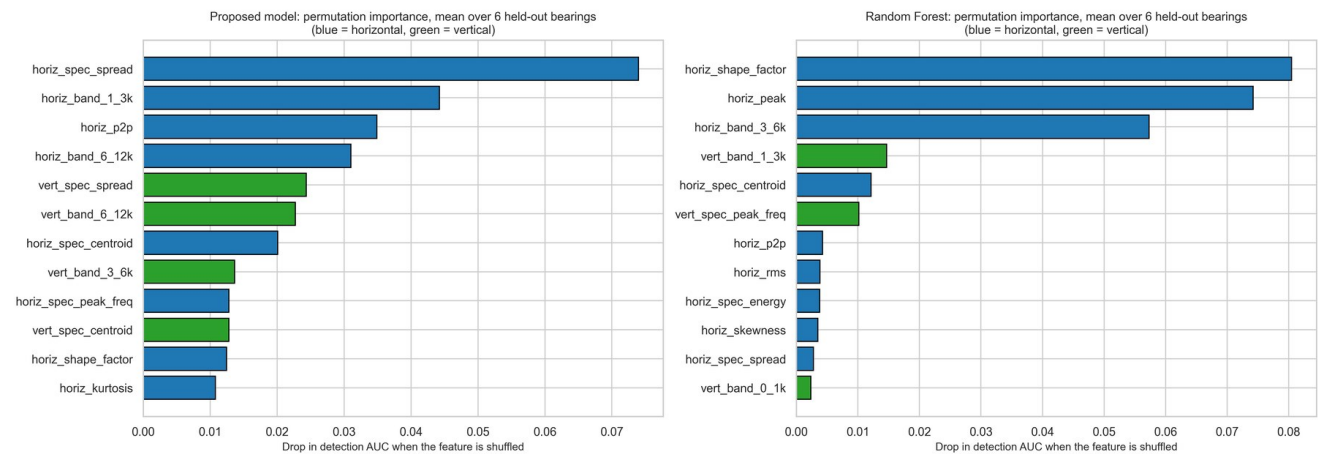


Figure G18. Permutation feature importance, proposed model versus Random Forest (held-out learning bearings). (Source: notebook cell [60]; saved file final_project/figures/fig_feature_importance_lobo.png)

What is shown	Top-12 features by the drop in detection AUC when that feature is shuffled across windows, averaged over the 6 folds; blue = horizontal, green = vertical channel.
How to read it	A longer bar means the model relies more on that feature. Importance is shared among strongly correlated features (12 feature pairs have $ p > 0.95$, cell [23]).
What the result shows	Proposed: horiz_spec_spread (+0.074), horiz_band_1_3k (+0.044), horiz_p2p (+0.035), horiz_band_6_12k (+0.031). Random Forest: horiz_shape_factor (+0.080), horiz_peak (+0.074), horiz_band_3_6k (+0.057). Rank agreement between the two models (Spearman) = -0.05 .
Reasonable conclusion	Both models rely mostly on horizontal-channel features, but the two models rely on different cues ; the importance is descriptive.
Do NOT conclude	That these features physically cause failure, or that they are “the” indicators: importance describes the model on this data, not the bearing physics.

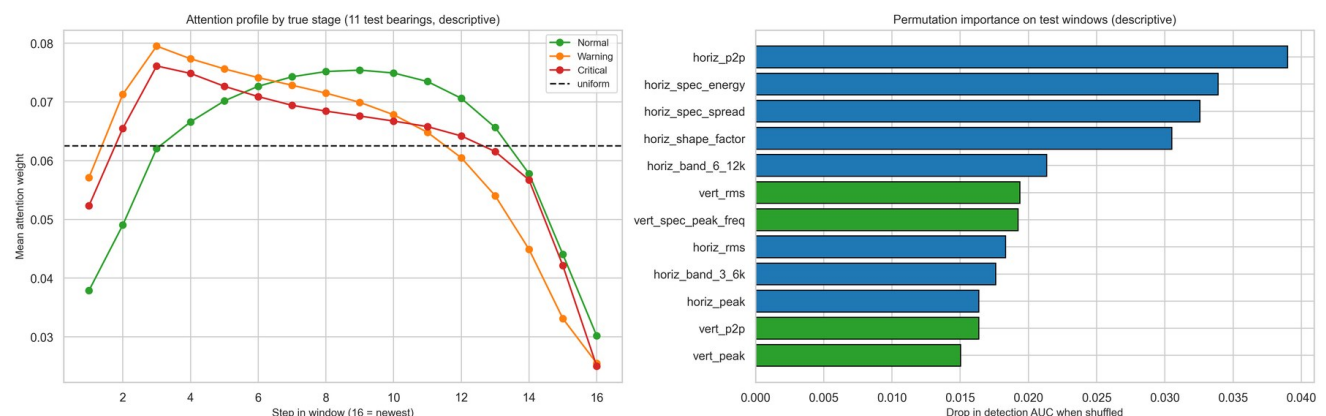


Figure G19. Descriptive explainability on the test bearings (no decision depends on it). (Source: notebook cell [74]; saved file final_project/figures/fig_explainability_test.png)

What is shown	Left: attention profile by true stage on the 11 test bearings. Right: permutation importance of the seed-42 final model on test windows.
How to read it	Same reading as for the validation version above.
What the result shows	Replacing learned attention by uniform weights changes the prediction by 2.5 min on average (correlation 0.990). Top features on test: horiz_p2p (+0.039), horiz_spec_energy (+0.034), horiz_spec_spread (+0.033), horiz_shape_factor (+0.031); 3 of the top-5 overlap with the LOBO top-5.
Reasonable conclusion	The conclusion about weak attention repeats on unseen bearings, and the feature ranking is only partly stable.
Do NOT conclude	That the explanation is validated or causal; it was computed after the design freeze and is purely descriptive.

6.8 Sample prediction from the saved model (cell [79], saved output)

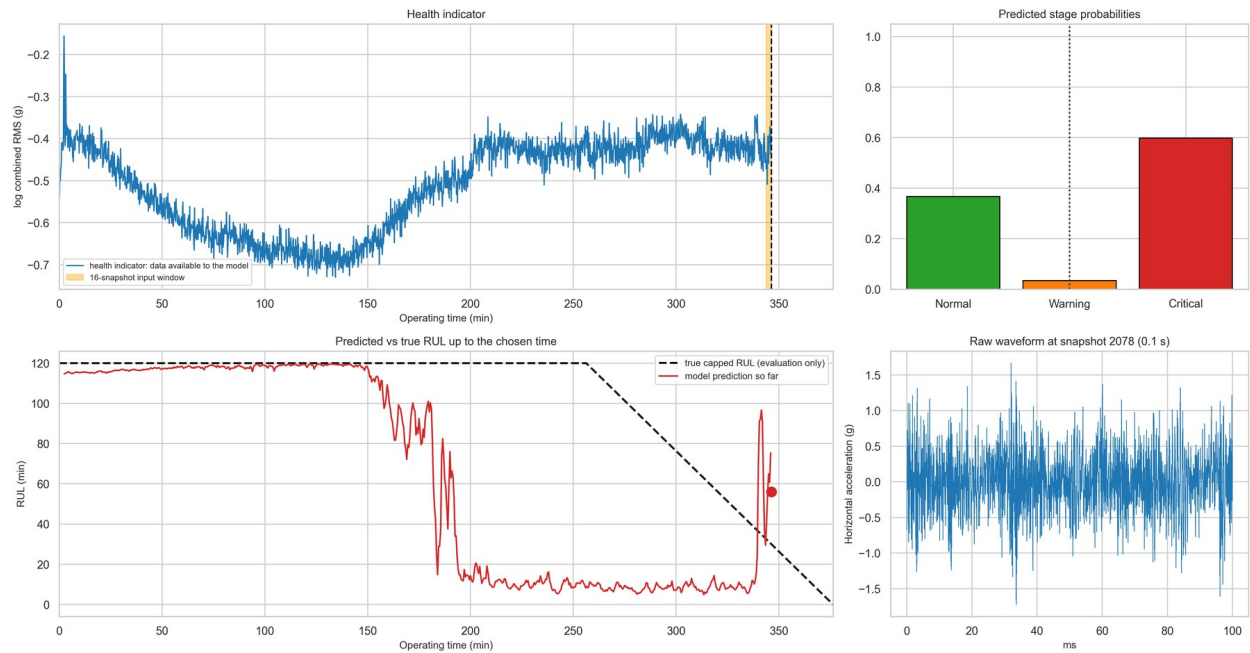


Figure G20. Demonstration on test Bearing1_7, 30 min before failure, using only data up to that time. (Source: notebook cell [79]; saved file final_project/figures/fig_sample_prediction_Bearing1_7_t346min.png)

What is shown	Health indicator with the 16-snapshot input window highlighted, stage probabilities, predicted versus true RUL up to the chosen time, and the raw 0.1 s waveform at that snapshot.
How to read it	The orange band is the data the model saw; everything to its right is hidden from the model.
What the result shows	Input window covers 343.8–346.3 min (snapshot 2078). Predicted RUL 56.0 min vs true 30.0 min (error +26.0 min). Stage head: Critical (N/W/C = 0.37/0.03/0.60); from predicted RUL: Warning; true stage: Warning.
Reasonable conclusion	The deployment path works end to end (causal window → saved scaler → saved network → RUL and stage) and the prediction is in the right direction but 26 min too optimistic.
Do NOT conclude	That this single example is typical: the demo bearing was chosen after the test result as the median actionable-error bearing (disclosed in cell [78]); a second example (cell [80], healthy phase at 94.0 min) predicted 115.8 min vs true capped 120.0 (Normal, probability 1.00).

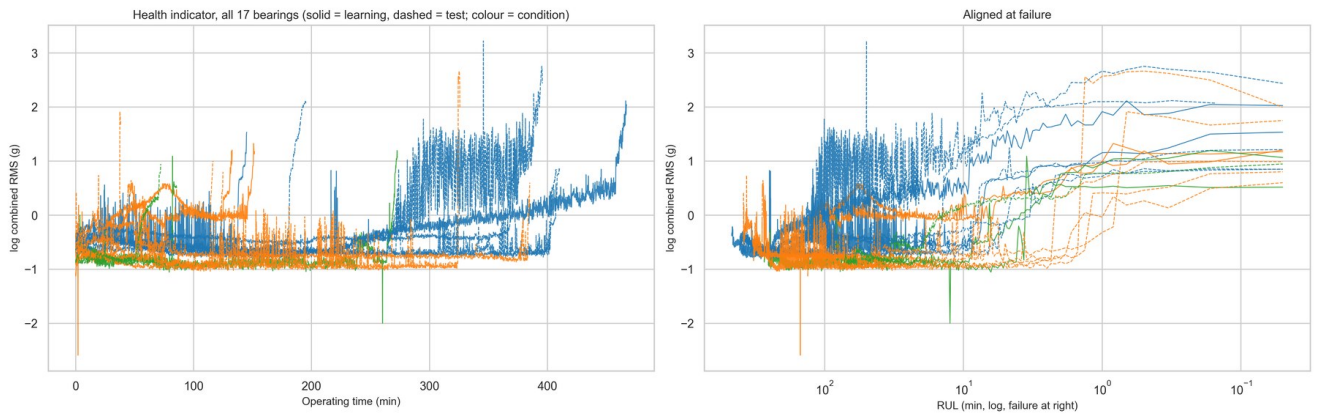


Figure G21. Health indicator of all 17 bearings (shown only after the design freeze). (Source: notebook cell [74]; saved file final_project/figures/fig_health_indicator_all17.png)

What is shown	Left: indicator versus time (solid = learning, dashed = test). Right: aligned at failure on a log axis.
How to read it	As the learning-bearing figure in 6.1 (cell [14]), but now with the test bearings added for description.
What the result shows	Test bearings also differ widely: some rise gradually, others only in the last minutes (descriptive phase table in cell [74]: e.g. Bearing2_3 final phase 1.3 min, Bearing1_7 175 min at the 10 % threshold).
Reasonable conclusion	The abrupt-versus-gradual split explains most of the per-bearing performance differences (e.g. proposed AUC@10 = 0.99 on 1_3 and 1_4, but 0.44 on 2_4).
Do NOT conclude	That phase length is a label or predictor: it is hindsight-based and was never used for decisions.

7. Metrics explained

For each metric: full name, what it measures (beginner English), direction, the actual value in this project, the evaluation population, units and limitation. All values are from the final test (11 unseen bearings) unless stated otherwise.

Metric (full name)	What it measures	Better =	Actual value (proposed, test)	Source / limitation
MAE – Mean Absolute Error (min)	Average size of the RUL mistake, in minutes, against the <i>capped</i> true RUL, over all windows.	lower	29.592 min (± 5.308 , 3 seeds); constant 35.623	[69]. Dominated by the healthy phase, where a constant is hard to beat.
Actionable-region MAE (min)	Same error but only for windows with true RUL ≤ 60 min, the zone where a prediction is useful.	lower	38.072 min (± 6.187); constant 56.354; best GB 32.651	[69]. Only computed where failure is within 60 min.
RMSE – Root Mean Squared Error (min)	Like MAE but punishes big mistakes more.	lower	44.11 (mean of seeds 38.74 / 45.36 / 48.24)	final_metrics_per_model_seed.csv; not shown in any notebook output.
AUC@H – Area Under the ROC Curve for ‘failure within H min’	Probability that a window truly closer to failure gets a higher alarm score ($\sim$ predicted RUL) than a healthier one. 0.5 = coin flip, 1 = perfect.	higher	AUC@10 0.759; AUC@60 0.749; mean over H = 5,10,20,30,60: 0.749 ; constant 0.500	[69], [76]. A detection score, not an RUL accuracy.
Accuracy (stage)	Fraction of windows with the correct stage.	higher	0.671 (uniform rule); 0.596 (native stage head)	[71]. Inflated by the large Normal class (constant gets 0.774!).
Precision / Recall (per class)	Precision: of windows called class X, how many truly are X. Recall: of windows truly X, how many were found.	higher	Uniform rule: recall N/W/C 0.764/0.328/0.391; precision 0.867/0.312/0.255	[71]. Critical recall is the operationally important one.
Macro-F1 (stage)	Average over the 3 classes of $F1 = 2PR/(P+R)$; treats classes equally.	higher	0.468 (uniform rule; ± 0.045); native head 0.377; constant 0.291	[69], [71].
False-alarm rate	Fraction of truly Normal windows (RUL > 60 min) that lie inside an alarm (3 consecutive windows predicted $\geq$ Warning).	lower	0.172 (± 0.109); seed 42 only: 0.053	[69]. A constant has 0 because it never alarms. Large seed spread.
Alarm active at end (of 11)	Number of test bearings whose alarm is still on at the last window before failure (strict).	higher	8.3 of 11 (seed mean)	[69]. Does not say how early the alarm began.
Sustained lead time (min)	Remaining life at the start of the final uninterrupted alarm run (capped at 120).	higher (with low false alarms)	16.32 min (median over seeds of the per-seed median lead)	[72]. A long run may be partly false alarm; read it with the false-alarm rate.
PHM-2012 official score	One prediction per test bearing at the truncation point; asymmetric exponential score (late predictions punished more); mean over 11 bearings.	higher (max 1)	0.146 $\pm$ 0.088 (3-seed mean); seed 42: 0.097; constant 0.124	[69]. 11 numbers only; capped predictor; not comparable to leaderboards.

7.1 Equations

Mean absolute error and F1:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad F_1 = \frac{2PR}{P+R}$$

Official PHM-2012 score for one bearing (cell [69], src/metrics.py: phm_score_single):

$$\%Er_i = 100 \frac{\text{Act}_i - \text{Pred}_i}{\text{Act}_i}$$

$$A_i = \exp\left(-\ln(0.5) \frac{\%Er_i}{5}\right) \text{ if } \%Er_i \leq 0 \text{ (late);} \quad A_i = \exp\left(+\ln(0.5) \frac{\%Er_i}{20}\right) \text{ if } \%Er_i > 0 \text{ (early)}$$

$$\text{Score} = \frac{1}{11} \sum_{i=1}^{11} A_i$$

In words: a perfect prediction gives $A = 1$; being 50 % too early gives about 0.18; being 5 % too late gives 0.5 (late is penalised four times more steeply than early).

7.2 Regression versus classification versus prognostic-challenge scores

Family	Question it answers	Metrics here	Typical trap
Regression	How many minutes off is the RUL?	MAE, actionable MAE, RMSE	Overall MAE hides poor behaviour near failure because most windows are healthy.
Classification / detection	Is the bearing in a given stage? Is failure within H minutes?	Accuracy, precision, recall, macro-F1, AUC@H, false-alarm rate	Accuracy is inflated by the majority class; AUC says nothing about the size of the RUL error.
Prognostic challenge score	Is the single RUL estimate at the truncation point accurate, with late errors penalised more?	PHM-2012 score A	Only 11 numbers; a capped predictor cannot match RUL > 120 min.

Measured result versus hypothesis. A *measured* result is a number produced by the saved experiment (e.g. test AUC 0.749). A *hypothesis* is a claim you believe but have not shown (e.g. “attention improves accuracy”, “every bearing can be warned”). Present measured numbers as facts, and hypotheses as hypotheses; where the notebook tested a hypothesis and it was not supported (H2 mixed, H3 not supported), say so.

8. Selected notebook cells for the live review

Recommendation: 13 cells, about 9.75 minutes. This is the smallest set that covers the problem, data, leakage control, model, results, honest limits and explainability. The notebook has 93 cells (zero-based [0]–[92]); you do **not** need to demonstrate them all.

Rules for the live review. (1) Do not press Run All or run any cell: all outputs are already saved; scroll to the cell and show the saved output. (2) The notebook file is 7.8 MB; if the viewer is slow, open the same figure from `final_project/figures/` or the table from `final_project/artifacts/` (saved report artifacts). (3) Cell [79]’s ‘EDIT THESE TWO LINES’ demo needs a run to change the bearing; use the saved output instead.

#	Cell	Section heading	What to display	Why it matters	What to say aloud	Min
1	[1]	1 Research questions	Markdown cell [1] (nothing to run). Do not start at cell [0]: its abstract is the literal text ABSTRACT_PLACEHOLDER.	States the six RQs and three hypotheses H1–H3 up front.	Title and goal; “I will show which RQs the evidence answers and where it does not.”	0.5
2	[8]	5 Dataset audit – Fig. 1	Saved figure output of cell [8] (if slow, open <code>final_project/figures/fig01_dataset_overview.png</code>).	17 bearings, 3 conditions, lifetimes 38–467 min.	PRONOSTIA/PHM-2012, 24,889 snapshots, 25.6 kHz, 6 learning + 11 test, lifetimes differ $\times 12$.	0.75
3	[12]	6 Raw data – Fig. 3	Saved figure output of cell [12].	Healthy versus end-of-life snapshot, waveform and spectrum.	Vibration in g, impacts and broad spectrum at end of life → motivates the 34 features.	0.75
4	[14]	7 Degradation – Fig. 4 + phase table	Saved output of cell [14] (figure + table).	Shows gradual versus abrupt degradation: the heart of the difficulty.	Bearing1_1 slow; 3_1, 3_2, 1_2 flat until last 4–9 min → no model can warn early there (data property).	0.75
5	[35]	13 Split protocol + leakage demo	Saved output of cell [35]: fold table, split figure and the 2-row table (1.5 vs 42.2 min).	Why bearing-level splitting; quantified leakage.	Train/validation/test per bearing; random split gives fake MAE 1.5 min and AUC 1.00; correct split MAE 42.2, AUC 0.70.	1.0
6	[39]	15 Architecture	Saved output of cell [39] (layer table + diagram).	The verified model: CNN → BiLSTM → attention → two heads; 114,692 parameters.	Explain each block simply; mention 6 learning bearings so over-fitting risk.	1.0
7	[42]	16 Training curves	Saved figure output of cell [42].	Training vs validation loss.	Training loss → ~0, validation rises: memorisation; data is the bottleneck; early stopping and 3 final epochs.	0.5
8	[69]	22 Final test metrics + PHM score	Saved output of cell [69] (11-model table and per-bearing PHM table).	The one-shot test result for all models and the official score.	AUC 0.749 vs 0.50; actionable MAE 38.1 vs 56.4; but tied with CNN / dual-head and GB is lower in MAE; PHM 0.146 ± 0.088 vs constant 0.124.	1.0
9	[70]	22 Test figures	Saved figures of cell [70]; start with the 12-panel per-bearing figure.	Predicted vs true RUL for every test bearing; scatter, ROC, confusion.	Good on gradual bearings (1_3, 1_4), no warning on abrupt ones (2_3, 2_7), long false alarm on 1_7.	0.75
10	[72]	22 Per-bearing table + early warning	Saved output of cell [72] (table + timeline figure).	Per-bearing AUC, false alarms and sustained lead time.	Useful warning for only 3/11 bearings (1_3, 1_4, 3_3) in seed 42.	0.5
11	[59]	20 Explainability	Saved figure output of cell [59].	Attention profile and uniform-attention test.	Attention near uniform; replacing it changes prediction by 1.4 min: inspectable, not causal.	0.5
12	[86]	26 Hypothesis verdicts	Saved Markdown output of cell [86].	Code-computed verdicts for H1–H3 and RQ3–RQ6.	H1 partly supported; H2 mixed; H3 not supported.	0.75

#	Cell	Section heading	What to display	Why it matters	What to say aloud	Min
13	[84]	25 Limitations (+ conclusion cell [87])	Markdown cell [84]; optionally read the one-paragraph conclusion in cell [87].	Honest limits and what is not claimed.	Six bearings, abrupt failures, label uncertainty, prior exposure, attention not causal; future work.	0.5

Total $\approx$ 9.75 min. If you have only 5 minutes, use order 2, 5, 6, 8, 12. If the guide wants more, use the fallback list.

8.1 Optional cells for follow-up questions

Cell	Use it when the guide asks...
[7]	Full 17-bearing audit table + anomalies (Bearing1_4)
[11]	Raw file table and plot (Fig. 2)
[17]	Feature screening table + Fig. 5
[21]	34-feature documentation table
[23]	Feature distributions and correlation (Figs. 6–7)
[26]	Target-formulation probe table (A, A', B, C)
[27]	Horizon detectability (Fig. 8)
[30]	Cap-sensitivity table
[32]	The 20 automated leakage checks (answer to “how do you know there is no leakage?”)
[45]	LOBO comparison tables (per-fold)
[47]	Model selection by LOBO (selection.json)
[51]	LOBO ablation table and figure
[55]	LOBO ROC / confusion / error distribution
[56]	LOBO per-bearing table
[60]	Permutation importance
[61]	Explanation example for Bearing1_1
[64]	LOBO early-warning tables and timeline
[71]	Stage classification metrics on test (precision/recall)
[73]	Test-bearing ablation
[74]	Test explainability + 17-bearing health indicator + phase table
[79]	Sample prediction 30 min before failure (saved output, do not re-run)
[80]	Sample prediction in the healthy phase
[83]	Multi-criteria comparison table
[89]	Config and artifact list
[91]	The notebook's own 'How to explain this project' script

8.2 Where the notebook has no evidence (do not look for it)

- No window-length variation, no hyper-parameter search, no per-condition models, no uncertainty estimates (cells [84], [91]).
- No comparison with published PHM-2012 methods (not comparable: capped predictor; the notebook says so in [84]).
- No experiment on a second dataset.
- No saved training-history *numbers* for the final (all-6-bearings) models: the training curves in [42] are for LOBO folds only.

9. Complete speaking script (about 10 minutes)

Written in simple spoken English. Bracketed text like **[show cell 8]** tells you which saved output to show. Practise it aloud; each paragraph is roughly 40–60 seconds. Every number is from the saved outputs.

Introduction [cell 1]

Good morning, ma'am. Our project is about *bearing health monitoring*. A bearing lets a shaft rotate smoothly in motors and pumps. When it fails suddenly, the machine stops and that is expensive. Our aim is to use vibration data to estimate how much life a bearing has left, called remaining useful life or RUL, and to raise a Normal, Warning or Critical alarm. The title is Explainable Dual-Head CNN-BiLSTM-Attention Prognostics. We asked six research questions about the data, the target, generalisation, the architecture, explainability and limits, and I will show you what the evidence says for each.

Problem statement

In simple words: given the last few minutes of vibration of a bearing, predict how many minutes it can still run and which stage it is in, and do this for a bearing the model has *never seen*. That last part is important, because a real maintenance engineer will always meet a new machine.

Dataset [cell 8, cell 12]

We used the public PRONOSTIA dataset from the IEEE PHM 2012 challenge. There are 17 bearings run until failure: 6 for learning and 11 for testing, under three operating conditions of speed and load. Two accelerometers, horizontal and vertical, record at 25.6 kilohertz for one tenth of a second every 10 seconds, so we have 24,889 snapshots. Lifetimes range from 38 to 467 minutes, a factor of about 12. This figure shows a healthy snapshot and a snapshot at the end of life: at the end the vibration has strong impacts and its frequency content is much broader. This is why we compute amplitude and frequency features.

Degradation [cell 14]

This plot is very important. It shows a simple health indicator for the six learning bearings. Bearing1_1 degrades slowly over hours, but Bearing3_1, 3_2 and 1_2 look almost flat until the last four to nine minutes. So some bearings give almost no warning in the vibration. That is a property of the data, not a mistake in the model.

Methodology: features and targets

Each 0.1-second snapshot is converted to 34 features: 17 per channel, covering RMS, peak, kurtosis, crest factor, spectral centroid, spectral spread and energy in four frequency bands. The input of the model is the last 16 snapshots. For the target we use the remaining life capped at 120 minutes, because a bearing that is healthy gives no evidence about whether 200 or 400 minutes remain. We also use three stages: Normal above 60 minutes, Warning between 20 and 60, and Critical below 20. These numbers are design choices fixed in advance, and we checked that the conclusion does not depend on the cap.

Leakage and the split [cell 35]

To avoid data leakage we split by whole bearings, not by random windows. Neighbouring windows share 15 of 16 snapshots, so a random split lets the model see near copies of its test data. We measured it: with the same Random Forest, a random split gives an error of 1.5 minutes and an AUC of 1.00, while the correct bearing-level split gives 42.2 minutes and 0.70. The six learning bearings are used in leave-one-bearing-out validation, and the 11 test bearings are scored only once at the end. We also wrote 20 automatic checks, such as a poison test and a truncation test, and all passed.

Model architecture [cell 39]

The model takes a 16 by 34 window. Two convolution layers find short local patterns. A bidirectional LSTM follows the trend inside the window; it is bidirectional only inside the past window, so it never sees the future. An attention layer gives a weight to each of the 16 time steps. Then two heads: one outputs the RUL in minutes, the other outputs the probabilities of the three stages. The model has 114,692 parameters, which is a lot for six learning bearings, so we expected over-fitting.

Training [cell 42]

This is exactly what the training curves show. The training loss drops to almost zero, but the validation loss on an unseen bearing stays high or rises. The model memorises the training bearings. So we used early stopping and trained the final model for only three epochs. For us the bottleneck is the amount of data, not the size of the network.

Results [cell 69, cell 70, cell 72]

On the 11 unseen test bearings, the proposed model detects an approaching failure better than chance: the average AUC is 0.75, where 0.50 is a coin flip. The error of the remaining-life estimate in the last 60 minutes is 38 minutes, compared with 56 for a constant guess. But I must be honest. A plain CNN and a dual-head model without attention are statistically tied with our model, and Gradient Boosting has a lower error. The official PHM score is 0.146 with a seed spread of 0.088, versus 0.124 for a constant predictor, so we do not claim an improvement there. The per-bearing figure shows why: for gradual bearings such as 1_3 and 1_4 the prediction follows the real RUL, but for abrupt bearings such as 2_3 and 2_7 the model predicts a healthy bearing until the end. A useful warning appears for only three of the eleven bearings.

Explainability [cell 59]

For explainability we looked at attention weights and permutation importance. The attention weights are close to uniform, and if we replace the learned attention by equal weights the prediction changes only by 1.4 minutes on average. So attention is inspectable, but it is functionally weak and it is not a causal explanation of the failure. Permutation importance shows that the model mostly uses horizontal-channel features, but the proposed model and the Random Forest rely on different features.

Hypotheses and limitations [cell 86, cell 84]

We tested three hypotheses. H1, that sequence models detect failure better than tabular models, is partly supported: it holds on validation but not clearly on test. H2, that attention and the dual head improve accuracy, is mixed, so we cannot claim it. H3, that every bearing gets a warning tens of minutes before failure, is not supported. The main limits are: only six learning bearings, abrupt failures with almost no precursor, uncertainty in the end-of-life labels, and that we looked at test-bearing curves in earlier experiments, which we disclose.

Conclusion

To conclude: we built a leakage-controlled, fully audited evaluation of a dual-head CNN-BiLSTM-Attention model on real bearing data. The model recognises imminent failure better than chance, but accurate minute-level RUL and long warnings for every bearing are not supported by this dataset. Our main contribution is the honest, bearing-level evaluation and the clear statement of what the data can and cannot show. Future work is more data, per-condition models, uncertainty estimates and a second dataset. Thank you, I am happy to take questions.

If you get lost: say “the honest summary is: it detects imminent failure better than chance, it is not clearly better than simple baselines, and early warning is not possible for abrupt bearings.” That sentence is supported by cells [69], [76], [86].

10. Faculty viva questions and answers

50 questions in 9 groups. Answers are grounded in cells and artifacts; where the notebook does not establish a fact, the answer says how to respond honestly. The four hardest follow-ups are marked [HARD].

A. Basic project understanding

Q1. What is your project in one sentence?

We use vibration data from real run-to-failure bearings to estimate remaining useful life and a Normal/Warning/Critical stage with a dual-head CNN-BiLSTM-Attention network, and we evaluate it strictly on bearings it has never seen (cell [91]).

Q2. Why does this problem matter?

Bearing failures stop machines unexpectedly and are expensive; predicting how close a bearing is to failure lets maintenance be planned (cell [91], item 1).

Q3. What is the main result?

On 11 unseen bearings the model detects imminent failure better than chance (mean AUC 0.749 vs 0.50) and has lower actionable-region RUL error than a constant (38.1 vs 56.4 min), but it is not clearly better than simple baselines and does not give long warnings for abrupt bearings (cells [69], [76], [86]).

Q4. What are your actual contributions?

A leakage-controlled bearing-level protocol with 20 passing checks, a measured demonstration of the leakage inflation (1.5 vs 42.2 min), a dataset audit that found the Bearing1_4 anomaly, and an honest comparison of 11 models with ablations. The architecture itself is a combination of standard components.

Q5. Is this a deployed real-time system?

No. It is an offline study on a public benchmark. A saved model, scaler and feature configuration exist and a causal one-window demo (cell [79]) shows how deployment would work, but no real-time system was built.

B. Dataset and vibration signals

Q6. Which dataset did you use and how big is it?

PRONOSTIA / IEEE PHM 2012: 17 bearings (6 learning, 11 test), 24,889 snapshot files, each 2560×6 (two accelerometer channels), 127,431,680 raw samples in total (cell [7]).

Q7. What do the two channels and the sampling rate mean?

Two accelerometers 90° apart measure horizontal and vertical vibration in g. Sampling is 25.6 kHz, so a 2560-sample snapshot lasts 0.1 s; one snapshot is taken every 10 s (cell [2] facts 7–10; time base verified in cell [11]).

Q8. How do you know when a bearing has failed?

Not by thresholding the snapshots: the 20 g criterion is not visible in every 0.1 s snapshot (Bearing1_5 peaks at 14.1 g; Bearing1_2 exceeds 20 g at 14.6 % of life). End of life is the official recording duration for learning bearings and truncation point + official RUL for test bearings (cells [7], [9]).

Q9. Did you find any problem in the dataset?

Yes. In Full_Test_Set, Bearing1_4 contains 289 snapshots after the truncation point = 2890 s but the official RUL is 339 s, so 2551 s lie after the official end of life. We truncate at the official end of life. Only Bearing1_4 uses ';' as delimiter. Test_set and Full_Test_Set are numerically identical where they overlap (33/33) (cell [7]).

Q10. Why do lifetimes differ so much and why does that matter?

Lifetimes are 38–467 min (factor ≈ 12) and degradation can be gradual or abrupt. It means a single threshold or a lifetime-normalised target cannot be used and that six learning bearings give a very small sample (cells [8], [9], [15]).

C. Feature engineering and preprocessing

Q11. Why 34 features instead of the raw signal?

2560 raw samples per snapshot would need far more data than six bearings can provide. 17 documented features per channel (9 time-domain, 4 spectral shape, 4 band-energy) give compact, physically meaningful inputs (cell [20]).

Q12. Why the log transforms? Are they fitted on the data?

Heavy-tailed features would dominate a neural network. The transforms ($\log_{10}$, $\log_{10}(x+1e-6)$, signed $\log_{1p}$) are fixed and non-fitted, chosen from distribution diagnostics, so they cannot leak. Maximum |skewness| falls from 16.1 to 5.9 (cell [23]).

Q13. Did you select features?

No. Screening on the learning bearings (cell [17]) is descriptive; all 34 features are always used. Screening showed that no single feature is a consistent health indicator across bearings (only 2 of 34 with consistency > 0.5).

Q14. How is scaling done without leakage?

A StandardScaler is fitted on the training bearings of each fold only (and on the 6 learning bearings for the final model) and then applied to validation/held-out/test windows; check 1 in cell [32] verifies this for every fold.

Q15. Many features are correlated. Is that a problem?

12 feature pairs have $|p| > 0.95$ (cell [23]). It is harmless for prediction but permutation importance is shared among correlated features, so importance must be interpreted with that in mind (cell [60]).

D. RUL formulation

Q16. Why is RUL capped at 120 minutes?

A healthy-looking bearing gives no evidence about whether 400 or 120 minutes remain, so uncapped targets contain unpredictable variance. 120 min is an a-priori operational planning horizon, not tuned; cell [30] shows the conclusion is the same for caps 60, 120 and 240.

Q17. Why not use the fraction of life (normalised RUL)?

It looked better (0.20 vs 0.25 MAE in the probe) only because it needs the bearing's total lifetime, i.e. future information that a running machine does not have. It is not deployable and was rejected (cell [29]).

Q18. How are the three stages defined? Were 20 and 60 min tuned?

Normal > 60 min, Warning 20–60, Critical ≤ 20 , computed from the true RUL only. They are design choices fixed before looking at results (cell [29], [32] config snapshot) and not tuned.

Q19. Why not train with a degradation-onset label from the vibration?

The onset is found with hindsight and depends on a threshold, so it cannot be a real-time claim. It is used only for description (phase-length tables) and never as a training label (cell [25] row D).

E. CNN, BiLSTM, attention and dual-head design

Q20. Why did you choose this architecture? [HARD]

Design arguments: CNN for local patterns and noise smoothing, BiLSTM for the trend inside the window, attention for a learned weighting that can be inspected, a second head to give an alarm level and possibly regularise (cell [38], [91]). These are arguments; the notebook **tested** them and found the benefit of attention and the dual head mixed (cell [86]). An honest answer is: the choice follows the project title and literature practice, and the evidence does not show it is better than simpler models.

Q21. The BiLSTM reads backwards. Does it use future data?

No. Backward reading is inside the 16-snapshot window, which lies entirely in the past of the prediction time; the truncation test in cell [32] confirms that deleting later snapshots does not change earlier windows.

Q22. What does attention add?

It gives one weight per time step (softmax, sum = 1) and the weighted sum is passed on. Functionally it adds little: replacing it with uniform weights changes the RUL prediction by 1.4 min on average (correlation 0.996) (cell [59]).

Q23. Why two heads, and did the second head help?

The stage head gives operators an alarm level and could regularise the shared representation. Results are mixed: adding the second head degraded actionable MAE on validation (+5.4 min, CI 0.07...12.6) but improved test AUC@10 (+0.065, CI 0.002...0.138) (cells [51], [73]).

Q24. 114,692 parameters for six bearings – isn't it too big?

Yes, the notebook says so (cell [39]) and the training curves confirm over-fitting (cell [42]). We used dropout, batch norm, early stopping on a validation bearing and only 3 final epochs. Capacity is not the bottleneck; data is.

Q25. Why Huber loss and these loss weights?

Huber ($\delta = 0.1$ on the 0–1 scaled target) is less sensitive to outliers than squared error. The weights (RUL 1.0, stage 0.5) were fixed a priori and not tuned. The notebook marks a citation for Huber loss as [CITATION NEEDED] (cell [92]).

F. Training, validation, testing and leakage**Q26. What is the difference between training, validation and test data here?**

Training bearings update the weights. A validation bearing is used only for early stopping and learning-rate reduction. The test bearings (11) are scored once, after the design is frozen. The unit is the bearing.

Q27. Why split by bearing? What if you split randomly? [HARD]

Windows of one bearing overlap by 15/16 snapshots, so a random split leaks near-duplicates. Measured: random split MAE 1.5 min / AUC 1.00 versus bearing-level 42.2 min / 0.70 (cell [35]).

Q28. How do you know there is no leakage?

20 executable checks passed (cell [32]): disjoint splits, scaler on training bearings only, poison test, truncation/causality test, window boundaries, independent feature recomputation, target-shuffle test, frozen config. In addition the audit shows selection.json was written before the test predictions. Caveat: the test bearings were looked at in earlier experiments (v1/v2), which motivated the capped-RUL formulation; we disclose this (cells [3], [84]).

Q29. Why did validation loss increase while training loss decreased?

Over-fitting: the model memorises four training bearings (stage accuracy gap 49 percentage points). We responded with early stopping and a short final training (cell [42], [43]).

Q30. How many epochs were used for the final model and why?

Three for the proposed model: the median best LOBO epoch was 1.0 but the rule is 'at least 3' (cell [47]); the final models train on all six learning bearings with fixed epochs because there is no validation bearing left.

Q31. Was the test set used for any decision?

No decision (model, epochs, cap, stage boundaries) used it. The demo bearing choice (median error) and the descriptive explainability were done after the test result and are disclosed as descriptive (cells [74], [78]).

G. Metrics and experimental results**Q32. What is MAE here and why is it large?**

Mean absolute error of the predicted capped RUL in minutes. Overall test MAE is 29.6 min; in the last 60 minutes it is 38.1 min. The model sees little change in vibration for most of a bearing's life and abrupt bearings give no warning, so exact RUL is not supported by the data (cell [29]).

Q33. What does AUC mean in your project?

The probability that a window truly closer to failure than H minutes receives a higher alarm score (–predicted RUL) than a healthier window. 0.5 is chance, 1 is perfect. Test mean over H = 5...60 min is 0.749 (cell [69]).

Q34. Does your model outperform the baselines? [HARD]

Not clearly. On test it has the highest mean detection AUC (0.749) but is statistically tied with CNN and the dual-head model without attention; Gradient Boosting has lower actionable MAE (32.7 vs 38.1 min). On LOBO validation the LSTM has higher AUC (0.886 vs 0.847). It beats the constant predictor on actionable MAE and AUC (cells [45], [76]).

Q35. Why was the proposed model not the model 'selected'?

The selection rules (fixed in advance, LOBO only) picked LSTM for detection and CNN+BiLSTM+Attention (single head) for actionable MAE; the proposed model was kept as 'interpretable by design' (cell [47]). The ranking on validation is not reproduced on test (rank correlations +0.47 and -0.52, cell [76]).

Q36. What is the official PHM-2012 score and what did you get?

One RUL prediction per test bearing at the truncation point with an asymmetric exponential penalty (late errors punished more), averaged over 11 bearings. Proposed 0.146 ± 0.088 (3 seeds; seed 42: 0.097); constant 0.124; Random Forest 0.078 (cell [69]). It is noisy, the predictor is capped at 120 min, and the result is not comparable to challenge leaderboards.

Q37. Your false-alarm rate is high. Why?

On validation the proposed model has 0.52 under the uniform rule; on test 0.17 (seed spread 0.05–0.27). The alarm uses the noisy predicted RUL, so alarms often start before the Warning zone; a long 'sustained lead' can therefore be partly false alarm (cells [66], [69]).

Q38. Are the differences statistically significant?

Only paired bootstrap 95 % confidence intervals over 6 (LOBO) or 11 (test) bearings were used; there is no multiple-comparison correction and seeds show noticeable spread (e.g. actionable MAE 32–44 min across seeds). Treat 'measurable' entries as exploratory (cells [51], [73]).

Q39. Is your result reproducible?

Seeds are fixed, TensorFlow deterministic ops are on, and the audit retrained the proposed model with seed 42 and reproduced the saved test predictions to $1.4e-14$ min (FINAL_AUDIT_automated.md check 16).

Q40. Why do your numbers differ from the earlier paper / README?

Earlier versions (v1/v2) used a different pipeline (e.g. 30 features, a 3-bearing test split, 113,924 parameters, MAE $\approx$ 93 min). The final notebook (34 features, LOBO + one-shot 11-bearing test, 114,692 parameters) supersedes them; quote only final-notebook numbers.

H. Explainability**Q41. Does attention explain why the bearing fails?**

No. Attention shows where the network put more weight inside the 160-s window; it is not causal [9]. We tested its function: replacing it by uniform weights changes predictions by 1.4 min (validation) and 2.5 min (test).

Q42. What does permutation importance show?

Shuffling one feature across windows and measuring the AUC drop. Proposed model: `horiz_spec_spread`, `horiz_band_1_3k`, `horiz_p2p`; Random Forest: `horiz_shape_factor`, `horiz_peak`, `horiz_band_3_6k`. It describes what the model relies on, not physical causes.

Q43. Why do the proposed model and Random Forest rely on different features?

Rank agreement is -0.05 (cell [60]). Features are strongly correlated so importance is shared, and with six training bearings the learned cues are not stable (only 3 of the top-5 features overlap between validation and test for the proposed model, cell [74]).

Q44. Can you call the model 'explainable'?

Only in a limited sense – inspectable. The title says 'Explainable', but the evidence supports inspectability (attention weights, importance), not causal explanation (cell [62]).

I. Limitations, novelty and future work

Q45. Is early warning shown for every bearing? [HARD]

No. H3 is NOT supported. A useful warning (alarm active at the end and sustained lead ≥ 10 min) exists for 3 of 11 test bearings (1_3, 1_4, 3_3 in seed 42): 2 of the 3 bearings with a visible degradation phase ≥ 20 min and 1 of the 8 abrupt ones (cell [86]). Bearings like 2_3 and 2_7 are predicted healthy until the end.

Q46. What is novel in your work?

The components (CNN, BiLSTM, attention, multi-task heads) are standard. The distinctive parts are the leakage-controlled and audited bearing-level evaluation, the measured effect of random splitting, and the honest component-wise ablation and negative results. Do not claim state-of-the-art accuracy.

Q47. What are the main limitations?

Six independent learning bearings; abrupt failures; label uncertainty (20 g not observable, Bearing1_4); design choices (cap, stage bounds) define success; prior exposure to test curves in earlier experiments; no hyper-parameter search; one rig and bearing type (cell [84]).

Q48. Why not compare with published PHM-2012 results?

They use different protocols; our predictor is capped at 120 min and the official score uses only 11 predictions, so the notebook states the result is not comparable with leaderboards. A peer-reviewed CNN-LSTM-attention PHM-2012 paper could not be verified and is marked [CITATION NEEDED] (cell [2]).

Q49. What would you do next?

Use leave-one-bearing-out over all 17 bearings for a deployment model, vary window length, per-condition models, uncertainty estimates, physically informed fault-frequency features, and a second bearing dataset (cell [91] item 16).

Q50. If you had more time or data, what would change most?

More independent bearings: the notebook identifies data, not capacity, as the bottleneck (cell [43]).

11. Research questions and limitations

11.1 Research questions and hypotheses versus evidence

Question / hypothesis	Evidence available	Supported finding	Unresolved	Status
RQ1 Data	Audit table [7], Fig. 1 [8], anomalies [9]	17 bearings, 24,889 clean files; lifetimes 38–467 min; Bearing1_4 over-runs its official EoL; 20 g not observable in every snapshot.	Label uncertainty remains (EoL from durations).	Answered
RQ2 Formulation	[25]–[30]	Capped RUL + stage chosen; life-fraction rejected (needs future); minute-level RUL not supported by RF probe; cap-insensitive conclusion.	Cap and stage bounds are design choices; no alternative multi-task weights tested.	Answered (with caveats)
RQ3 Generalisation	LOBO [45], test [69], [76]	Test AUC 0.749 (> 0.50); actionable MAE 38.1 vs 56.4 min; but PHM score 0.146 ± 0.088 vs constant 0.124 and large per-bearing variation.	Minute-level accuracy not shown; validation ranking not reproduced on test.	Partly answered
RQ4 Architecture	Ablations [51], [73]; baselines [45], [69]	Attention and dual head: MIXED; proposed tied with CNN / dual-head-no-attention on detection; GB lower RUL error.	No statistical power with 6/11 bearings; no multiple-comparison control; no window or width ablations.	Partly answered (benefit not shown)
RQ5 Explainability	[59]–[62], [74]	Model is inspectable; attention functionally weak (1.4 min change); permutation importance models disagree ($p = -0.05$).	No causal or physical validation of explanations.	Answered (negatively)
RQ6 Limits	[27], [14], [72], [74]	Early warning possible for gradual bearings (1_3, 1_4, 3_3 in seed 42) and not for abrupt ones.	Per-bearing results depend on seed (false alarm 0.05–0.27).	Answered
H1 sequence > tabular	[86]	LOBO: LSTM vs GB +0.162 AUC (CI +0.092...+0.238); test AUC@10 +0.048 (CI -0.097...+0.187).	Not confirmed on test.	PARTLY SUPPORTED
H2 attention and dual head improve accuracy	[51], [73], [86]	Attention 2 improvements/0 degradations/6 none; dual head 1/1/6.	No robust improvement.	MIXED – not supported
H3 warning for every bearing	[72], [86]	Useful warning for 3/11 test bearings (2/3 gradual, 1/8 abrupt).	Abrupt bearings give no early signal.	NOT SUPPORTED

11.2 Missing experiments and statistical caveats (only those that apply)

- **Baselines:** present (constant, Ridge, RF, Gradient Boosting, CNN, LSTM, CNN+LSTM, CNN+BiLSTM). **Missing:** published PHM-2012 methods, TCN/Transformer baselines, any model trained with more than 6 bearings.
- **Ablation:** present for CNN/LSTM/BiLSTM/attention/dual head. **Missing:** window length (fixed at 16), cap and loss-weight ablations on the network, per-condition models.
- **Statistics:** 2 seeds (LOBO) and 3 seeds (test); paired bootstrap over 6 or 11 bearings; no formal significance test across seeds, no multiple-comparison correction across the 35 ablation comparisons, seed spread is large for some metrics.
- **Dataset:** one rig, one bearing type, three conditions; condition 3 has two learning bearings (none in LOBO fold 4 training).
- **Reporting:** RMSE exists in CSV but is not shown in any cell output; the abstract in cell [0] is an unfilled placeholder.
- **Hindsight:** descriptive phase tables use whole trajectories; the capped-RUL formulation was motivated by earlier experiments that looked at test bearings.

11.3 Limitations list in the notebook (cell [84]) in one view

- Very few independent bearings (6 learning, 11 test).
- Heterogeneous operating conditions and degradation patterns.
- Abrupt failures (no model can warn early).
- Label uncertainty (20 g criterion, Bearing1_4, early level shifts in 2_1/2_2).
- A-priori design choices (cap 120, stage bounds 20/60).
- Hindsight in offline analysis; prior exposure to test bearings.
- Detection is not RUL estimation.
- RUL metric limits (healthy-dominated MAE; noisy PHM score).
- Attention is not a causal explanation.
- Small models, no hyper-parameter search, window not varied.
- CPU cost: about one hour for the whole notebook.
- Dataset specificity.

12. Final checklist

12.1 Cells to open before the review (saved outputs)

Open in this order: [1], [8], [12], [14], [35], [39], [42], [69], [70], [72], [59], [86], [84]. Keep these in tabs as backup: [32], [45], [47], [51], [73], [74], [79], [91].

12.2 Plots and tables to prepare (all already exist)

- figures/: fig01_dataset_overview, fig03_healthy_vs_degraded, fig04_health_indicator_learning, fig09_split_protocol, fig_architecture, fig_training_curves, fig_per_bearing_test, fig_confusion_test, fig_early_warning_test, fig_attention_lobo.
- artifacts/: final_model_comparison_test.csv, final_per_bearing_proposed.csv, ablation_test.csv, lobo_model_comparison.csv, selection.json, experiment_config.json.

12.3 Numbers to memorise

Quantity	Value	Cell
Bearings / snapshots / sampling	17 (6 learning + 11 test) / 24,889 / 25.6 kHz, 0.1 s every 10 s	[7]
Features / window / cap / stages	34 / 16 snapshots / 120 min / Normal > 60, Warning 20–60, Critical ≤ 20	[21], [29]
Parameters (proposed)	114,692	[39]
Random vs bearing split	MAE 1.5 vs 42.2 min; AUC 1.00 vs 0.70	[35]
Leakage checks	20 / 20 passed	[32]
Test AUC mean over H (proposed / constant)	0.749 / 0.500 (AUC@10 0.759)	[69]
Test actionable MAE (proposed / constant / best)	38.1 / 56.4 / 32.7 (Gradient Boosting) min	[69], [76]
Official PHM-2012 score	0.146 ± 0.088 (3 seeds); seed 42: 0.097; constant 0.124	[69]
Attention functional test	1.4 min change, correlation 0.996 (validation)	[59]
Useful warning	3 of 11 test bearings	[86]

12.4 Concepts to understand

- RUL, capped RUL, stages; bearing-level split vs window split; leave-one-bearing-out; scaler fitted on training only.
- CNN / BiLSTM / attention / dual head in your own words; why over-fitting happens with 6 bearings.
- AUC, MAE (overall vs actionable), macro-F1, false-alarm rate, PHM score; measured result vs hypothesis.
- Why attention weights are not causal explanations.

12.5 Questions to practise

The four [HARD] questions in Section 10 (architecture choice, random-vs-bearing split, baselines, early warning for all bearings), plus “why do your numbers differ from the earlier paper?”.

12.6 Claims you must avoid without evidence

- “The proposed model outperforms the baselines.” (Tied or beaten: [76], [86].)
- “We achieve accurate RUL prediction.” (Actionable MAE 38 min; PHM score 0.146 ± 0.088.)
- “Warnings are given tens of minutes ahead for every bearing.” (3/11.)
- “Attention explains the failure.” (Weak and non-causal.)
- “Attention and the dual head improve accuracy.” (MIXED.)
- “State-of-the-art” or comparison with challenge leaderboards. (Not comparable.)
- “No leakage is possible.” (Pipeline checks pass; prior exposure and selection effects remain.)
- Any number from the root-level v1/v2 reports (MAE ≈ 93 min, 113,924 parameters, 30 features, 3-bearing test) as a final result.
- “The test set was never seen.” (Say: never used for decisions; earlier experiments looked at test curves.)

13. Technical appendix

A. Environment and dependencies (cell [5] saved output)

Item	Value
Python	3.12.6
NumPy	2.2.6
pandas	2.2.2
scikit-learn	1.5.2
SciPy	1.14.1
TensorFlow (Keras 3)	2.20.0
Platform	Windows-11-10.0.22621-SP0
Other libraries imported (cell [5])	matplotlib, seaborn, IPython, joblib (cell [68])
Runtime	≈ 1 h on CPU, deterministic TensorFlow ops (README of final_project; cell [84])
Dataset path	ieee-phm-2012-data-challenge-dataset-master/ (Learning_set, Test_set, Full_Test_Set)

B. Important implementation details

- **Windows:** sliding window of 16 snapshots inside each bearing; label of the last snapshot (make_windows). Tabular models receive [last snapshot, window mean, last – first] (102 numbers) of the same window.
- **Baselines:** Constant (training mean / majority stage); Ridge($\alpha = 1$) + balanced logistic regression; Random Forest 200 trees, min_samples_leaf 5, balanced classifier; HistGradientBoosting 200 iterations, lr 0.05.
- **Alarm rule:** predicted stage $\geq$ Warning for 3 consecutive windows (same for every model); stage-from-RUL thresholds 60 / 20 min.
- **Constant and Ridge** are run with one seed only; neural models with 2 (LOBO) or 3 (final) seeds; the saved final_proposed_model.keras is the seed-42 model.
- **Saved artifacts** (20 files, cell [89]): features, predictions, metric tables, ablations, permutation importances, selection.json, experiment_config.json, final_proposed_model.keras (1.4 MB), final_scaler.pkl, final_feature_config.json.
- **Official score:** prediction at snapshot index (truncation – 1) of each test bearing, converted to seconds.

Attention layer (from src/models.py):

```
class TemporalAttention(layers.Layer):
    # alpha_t = softmax_t(u^T tanh(W h_t + b)); context = sum_t alpha_t h_t
    def call(self, x):
        score = tf.tensordot(tf.tanh(tf.tensordot(x, self.W, 1) + self.b), self.u, 1) # (B, T, 1)
        alpha = tf.nn.softmax(score, axis=1)
        return tf.reduce_sum(x * alpha, axis=1), alpha
```

Frozen configuration printed by cell [32] / saved in experiment_config.json:

```
WINDOW=16 RUL_CAP_MIN=120 STAGE_CRITICAL_MIN=20 STAGE_WARNING_MIN=60 ALARM_PERSISTENCE=3
BATCH=64 LR=1e-3 MAX_EPOCHS=40 PATIENCE=6 HUBER_DELTA=0.1
LOSS_WEIGHTS: rul_output=1.0, risk_output=0.5
SEEDS_LOBO=(42,43) SEEDS_FINAL=(42,43,44)
epochs_final_by_model: cnn 3, lstm 3, cnn_lstm 4, cnn_bilstm 3, cnn_bilstm_attn 6, cnn_bilstm_dual 3, proposed 3
```

C. Compact cell-index reference (all 93 cells, zero-based)

“Exec” is the Jupyter execution counter (1–49 in order: all code cells were run once, top to bottom). “Saved output” is read directly from the .ipynb JSON.

Cell	Type	Exec	Saved output	Section / purpose
[0]	md	–	–	Title + abstract (abstract text is the literal placeholder ‘ABSTRACT_PLACEHOLDER’)
[1]	md	–	–	Research questions RQ1–RQ6, hypotheses H1–H3

Cell	Type	Exec	Saved output	Section / purpose
[2]	md	–	–	Dataset facts and literature review (17 bearings, 25.6 kHz, EoL rule...)
[3]	md	–	–	Methodology decisions table; disclosed hindsight
[4]	md	–	–	Setup/reproducibility text
[5]	code	1	text	Imports, seeds, versions; extract features from all 24,889 raw files (98 s)
[6]	md	–	–	Dataset-audit text
[7]	code	2	markdown, table/result, text	Audit table of 17 bearings, assertions, Test_set vs Full_Test_Set check, anomaly list
[8]	code	3	figure, markdown	Fig. 1 dataset overview
[9]	md	–	–	Audit findings (3 key facts)
[10]	md	–	–	Raw-sample text
[11]	code	4	figure, markdown, table/result, text	One raw file: table + Fig. 2
[12]	code	5	figure, markdown	Fig. 3 healthy vs end-of-life (waveform + spectrum)
[13]	md	–	–	EDA text
[14]	code	6	figure, markdown, table/result, text	Build target table; Fig. 4 health indicator; final-phase length table
[15]	md	–	–	EDA findings
[16]	md	–	–	Health-indicator screening text
[17]	code	7	figure, markdown, table/result	Feature screening table + Fig. 5
[18]	md	–	–	Heading 8.1
[19]	code	8	markdown	Screening findings
[20]	md	–	–	Feature-engineering text
[21]	code	9	table/result, text	34-feature documentation table; saves feature_names.json
[22]	md	–	–	Feature-analysis text
[23]	code	10	figure, markdown, table/result, text	NaN check, skewness table, Fig. 6 distributions, Fig. 7 correlation
[24]	md	–	–	Findings 10.1
[25]	md	–	–	Target formulations A–E and decision principle
[26]	code	11	markdown, table/result	Random-Forest probe of formulations (table)
[27]	code	12	figure, markdown	Horizon detectability, Fig. 8
[28]	md	–	–	Heading 11.3
[29]	code	13	markdown	Frozen target decision text
[30]	code	14	markdown, table/result	Cap-sensitivity table
[31]	md	–	–	Leakage-check text
[32]	code	15	markdown, table/result	Run the 20 automated leakage checks
[33]	md	–	–	Findings 12.1
[34]	md	–	–	Split-protocol text
[35]	code	16	figure, markdown, table/result	LOBO fold table, Fig. 9, random-vs-bearing split table
[36]	md	–	–	Baseline-model text
[37]	code	17	text	Train classical baselines in LOBO
[38]	md	–	–	Architecture text

Cell	Type	Exec	Saved output	Section / purpose
[39]	code	18	figure, markdown, table/result, text	Build model; layer table; parameter counts; architecture figure
[40]	md	–	–	Training-recipe table
[41]	code	19	text	Train neural models in LOBO (1,567 s)
[42]	code	20	figure, markdown	Training/validation curves
[43]	md	–	–	Findings 16.1
[44]	md	–	–	Model-comparison text + metric definitions
[45]	code	21	markdown, table/result	LOBO comparison tables (+ per-fold)
[46]	code	22	figure, markdown	LOBO comparison bar figure
[47]	code	23	markdown	Model selection from LOBO → selection.json
[48]	code	24	markdown	Findings 17.1
[49]	md	–	–	17.2 hypotheses text
[50]	md	–	–	Ablation text
[51]	code	25	figure, markdown, table/result	LOBO ablation table + figure
[52]	code	26	markdown	Findings 18.1
[53]	md	–	–	Validation-evaluation text
[54]	code	27	figure, markdown	LOBO actual-vs-predicted and per-bearing trajectories
[55]	code	28	figure, markdown	LOBO ROC, confusion, error distribution
[56]	code	29	table/result	LOBO per-bearing table
[57]	md	–	–	Findings 19.1
[58]	md	–	–	Explainability text
[59]	code	30	figure, markdown	Attention profile + uniform-attention functional test
[60]	code	31	figure, markdown, text	Permutation importance (proposed vs RF)
[61]	code	32	figure, markdown	Explanation example, Bearing1_1
[62]	code	33	markdown	Findings 20.1
[63]	md	–	–	Early-warning text
[64]	code	34	figure, markdown, table/result	LOBO early-warning tables + timelines
[65]	md	–	–	Heading 21.1
[66]	code	35	markdown	Findings 21.1
[67]	md	–	–	Final-test text and guards
[68]	code	36	text	Final training on 6 bearings; one-shot scoring; save model/scaler
[69]	code	37	markdown, table/result, text	Test metric table + official PHM-2012 score
[70]	code	38	figure, markdown	Test figures (per bearing, scatter, ROC, confusion, errors)
[71]	code	39	markdown, table/result	Stage metrics on test (accuracy/precision/recall/F1)
[72]	code	40	figure, markdown, table/result	Per-bearing test table + early warning
[73]	code	41	figure, markdown, table/result	Test-bearing ablation
[74]	code	42	figure, markdown, table/result	Test explainability; 17-bearing health indicator; phase table
[75]	md	–	–	Heading 22.1

Cell	Type	Exec	Saved output	Section / purpose
[76]	code	43	markdown	Test findings text
[77]	md	–	–	22.2 interpretation
[78]	md	–	–	Demo text
[79]	code	44	figure, markdown	Demo: Bearing1_7, 30 min before failure
[80]	code	45	figure, markdown	Demo: healthy phase
[81]	md	–	–	Findings 23.1
[82]	md	–	–	Summary text
[83]	code	46	markdown, table/result	Multi-criteria table
[84]	md	–	–	Limitations
[85]	md	–	–	Heading 26
[86]	code	47	markdown	Hypothesis verdicts (code-computed)
[87]	md	–	–	One-paragraph conclusion
[88]	md	–	–	Heading 27
[89]	code	48	markdown, table/result, text	Config JSON, metrics JSON, artifact list
[90]	md	–	–	Heading 'How to explain'
[91]	code	49	markdown	Review-explanation script (30 s ... 10 questions)
[92]	md	–	–	References

D. Evidence files inspected and unresolved inconsistencies

File / folder	How it was used
final_project/final_bearing_prognostics_review.ipynb	Primary source: all 93 cells, sources and saved outputs read as JSON (not executed). SHA-256 recorded before and after: unchanged.
final_project/src/{config,data,leakage,experiment,models,metrics}.py	Read in full to verify splits, scaling, windows, targets, architecture, metrics. features.py, analysis.py, report.py were not read line by line (their behaviour is covered by cells [20]–[21], [32]).
final_project/artifacts/*.csv, *.json	Used for tables and cross-checks (final_metrics.json agrees with cell [69] output). final_proposed_model.keras and final_scaler.pkl exist but were not loaded (parameter count is taken from cell [39]).
final_project/figures/*.png	Existing saved figures used (down-sized copies embedded).
final_project/audits/FINAL_AUDIT.md, FINAL_AUDIT_automated.md	Independent audit: 16 automated checks PASS; selection.json precedes test predictions; retraining reproduces test predictions (max diff 1.4e-14 min).
final_project/README.md, PROJECT_INVENTORY.md	Project structure and run instructions (consistent with the notebook).
final_project/reports/*.md, scripts/*.py	Present; the research report is generated from the notebook (per README). Not used as a source for numbers.
Root: README.md, LICENSE (MIT), PAPER_RESULTS_PACKAGE.md, AUDIT_REPORT.md, REMEDIATION_REPORT.md, CHALLENGE_SINGLE_INSPECTION_SCORE.md, AUDIT_V2_REPORT.md, POST_REMEDIATION_VERIFICATION.md, ALL_11_TEST_BEARINGS_REPORT.md	All exist. First lines read only. They document earlier experiments (explainable_dual_head_bearing_prognostics.ipynb and the v2 notebook; 30 features, 3-bearing validation/test split, 113,924 trainable parameters, MAE ≈ 93 min). No number from them is used in this guide.
Root: models/ (dual_head_cnn_bilstm_attention.keras, scaler.pkl, feature_names.json)	Belongs to the earlier pipeline; not used.

Unresolved inconsistencies and observations

#	Observation	How to handle it
1	Cell [0] abstract is the literal text 'ABSTRACT_PLACEHOLDER'.	Do not show cell [0]; use the 30-second summary in cell [91] (and Section 9 here).
2	Root-level reports and the README abstract describe an earlier pipeline whose numbers differ from the final notebook.	Quote only final-notebook numbers; if asked, say the final notebook supersedes v1/v2.
3	Bearing1_4 lifetime is 237.8 min in cells [7]/[8] (all files) but 195.2 min after truncation at its official end of life (cell [74]).	Both are correct for different populations.
4	H3 text in cell [86] reports a false-alarm rate of 0.05 (seed 42, mean over bearings) while cell [69] reports 0.172 (mean over 3 seeds).	Different aggregation; false alarm varies 0.05–0.27 by seed.
5	Frozen config prints SEED_LOBO = 42 although LOBO used seeds 42 and 43 (SEEDS_LOBO).	Cosmetic; experiment_config.json lists both seeds.
6	fit_scaler contains a vacuous assert ('... or True').	Learning-only property is verified by check_splits / check_scaler_train_only in cell [32].
7	'160 s' window: 16 snapshots 10 s apart span 150 s between first and last.	Say '16 snapshots, about 160 s of history'.
8	Mean horizon AUC in cell [29] is not monotone (0.89 at 5 min, 0.64 at 20, 0.70 at 60).	Mention only as 'decreases with horizon, with noise'.
9	Cell [89] lists 35 PNG files; the project README says 34 final figures (one obsolete figure was archived).	Immaterial.
10	Constant and Ridge have one seed; their $\pm$ is not shown in cell [69].	Do not compare their 'variance' with neural models.
11	The demo bearing (Bearing1_7) was chosen after the test result as the median-error bearing.	Disclose it (cell [78] does).